

Digital Architecture Framework



Digital Architecture
Framework

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1 About Digital Architecture Framework (DAF)

The Digital Architecture Framework (DAF), formerly known as "Delta Architectural Framework," is a set of modeling tools for Digital Engineering aiming to enable collaboration between roles such as Enterprise Architects, Solution Architects, and Business Analysts in designing the enterprise by creating high-quality models that can be transformed into artifacts such as code and documentation.

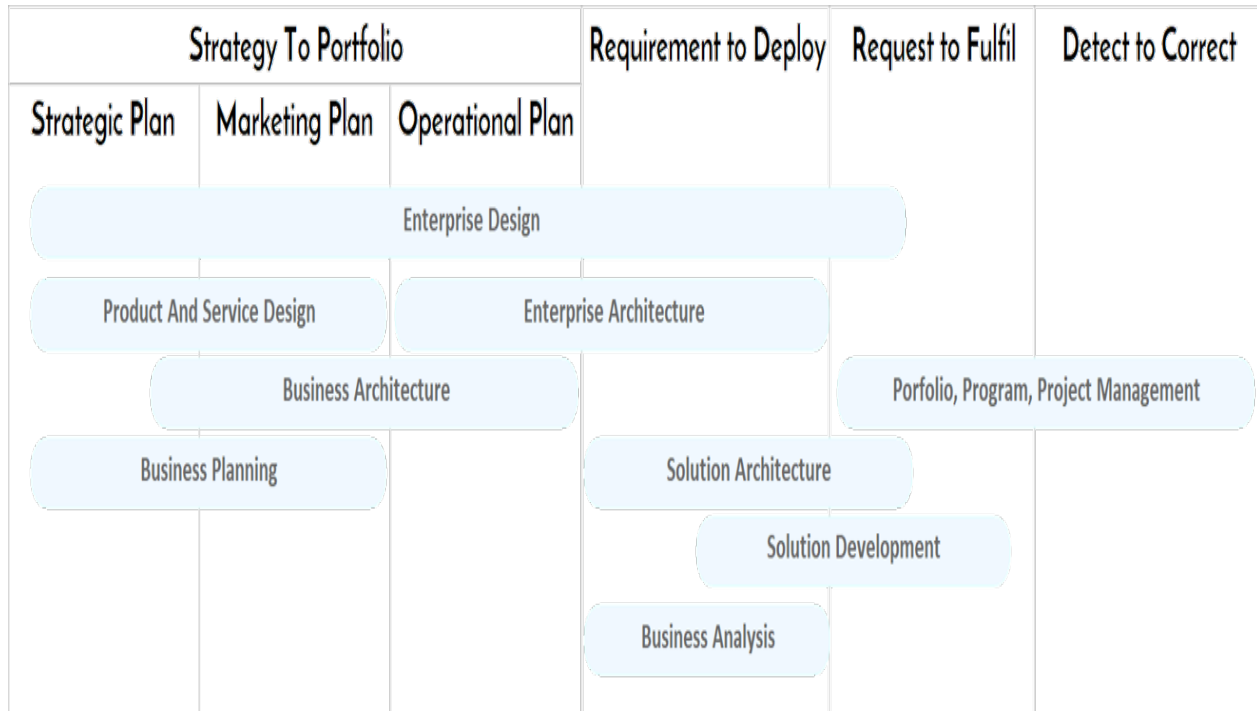


Figure 1: Value Streams covered by DAF

The diagram shows the classic gaps between strategy, design, and execution. DAF may accelerate the delivery of projects by supporting collaboration of different roles and by creating traceability between strategy and execution. In fact, it provides the metamodel to close these gaps, ensuring that strategic plans are directly reflected in architectures, architectures are transformed into solutions, and solutions generate tangible deliverables with traceability back to strategy.

The DAF meta-model is based on industry standards such as UML, ArchiMate, TOGAF, RUP, SPEM and best practices for requirement management, and offers a baseline of more than 50 concepts and properties that can be customized to meet specific needs. While DAF can be used on its own, it is not intended to represent a new standard but rather serves as an accelerator that allows for quick tailoring of existing standards.

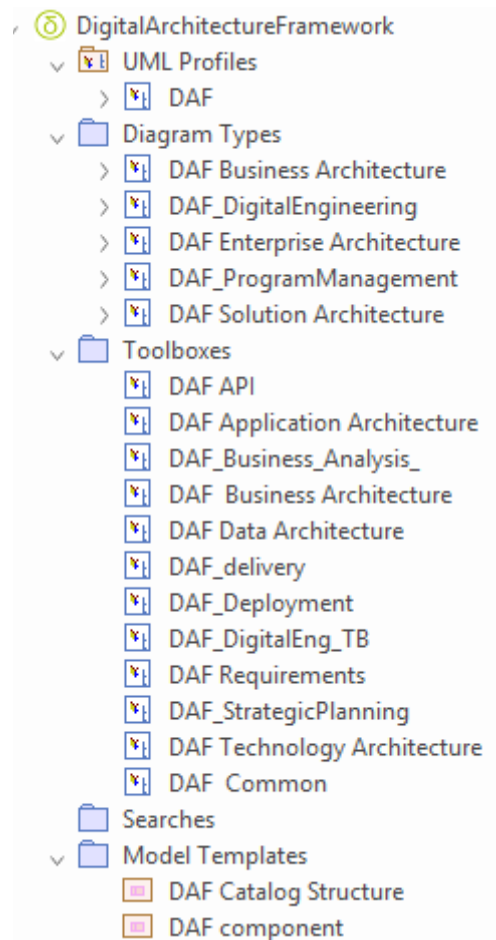


Figure 2: Content of the DAF MDG

DAF is currently implemented as MDG technology in Sparx Enterprise Architect, including:

- UML profile
- A set of customized icons
- Shape scripts
- multiple Toolboxes, organized by roles
- Custom diagrams, organized by roles
- Model patterns, accelerating the creation of DAF diagrams.
- Model Template with the structure of DAF Catalogs
- Quick linker
- A set of 100+ diagram examples showing how to use DAF to cover Enterprise Architecture, Solution Architecture, Design Architecture and Business Analysis. Those examples include all the diagrams in the TOGAF 9.1 specification.
- A structure based on TOGAF to host an enterprise repository
- A set of scripts, performing Model-to-Model (M2M) transformations
- A set of rules for model validation implemented in the EA Validator

2 License

- DAF is Open Source, released under the Creative Commons



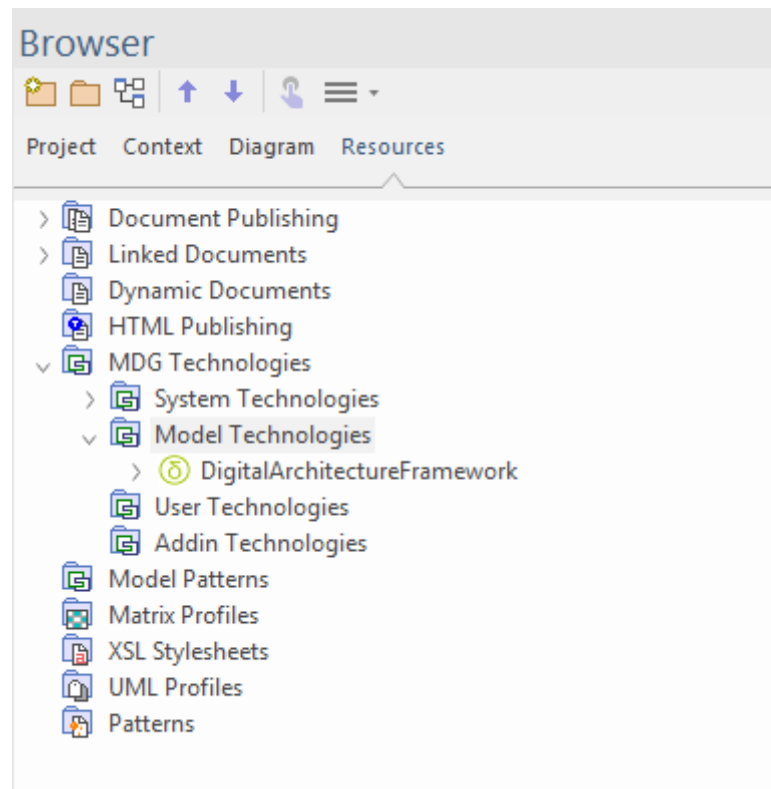
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3 Project

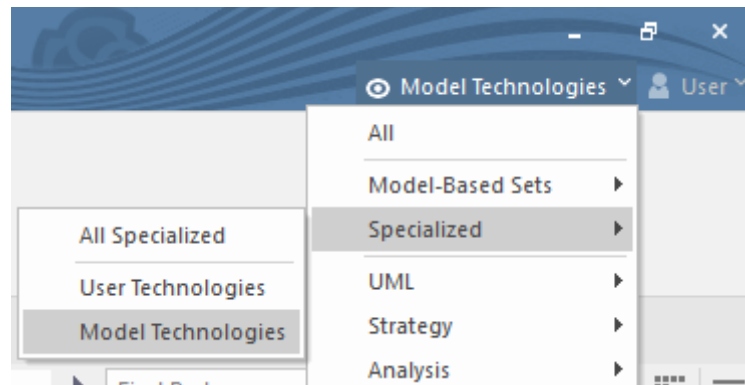
DAF is maintained in GitHub: [Digital Architecture Framework GitHub repository](#)

4 Installation

1. **Obtain the DAF MDG file:** Ensure you have the XML file defining the DAF MDG Technology from the repository [Digital Architecture Framework releases](#). This file contains the definitions for stereotypes, toolboxes, profiles, and diagram types.
2. Save the file to an accessible directory on your computer.
3. Launch **Sparx Enterprise Architect**.
4. Open or create a project where you want to use the DAF MDG Technology.

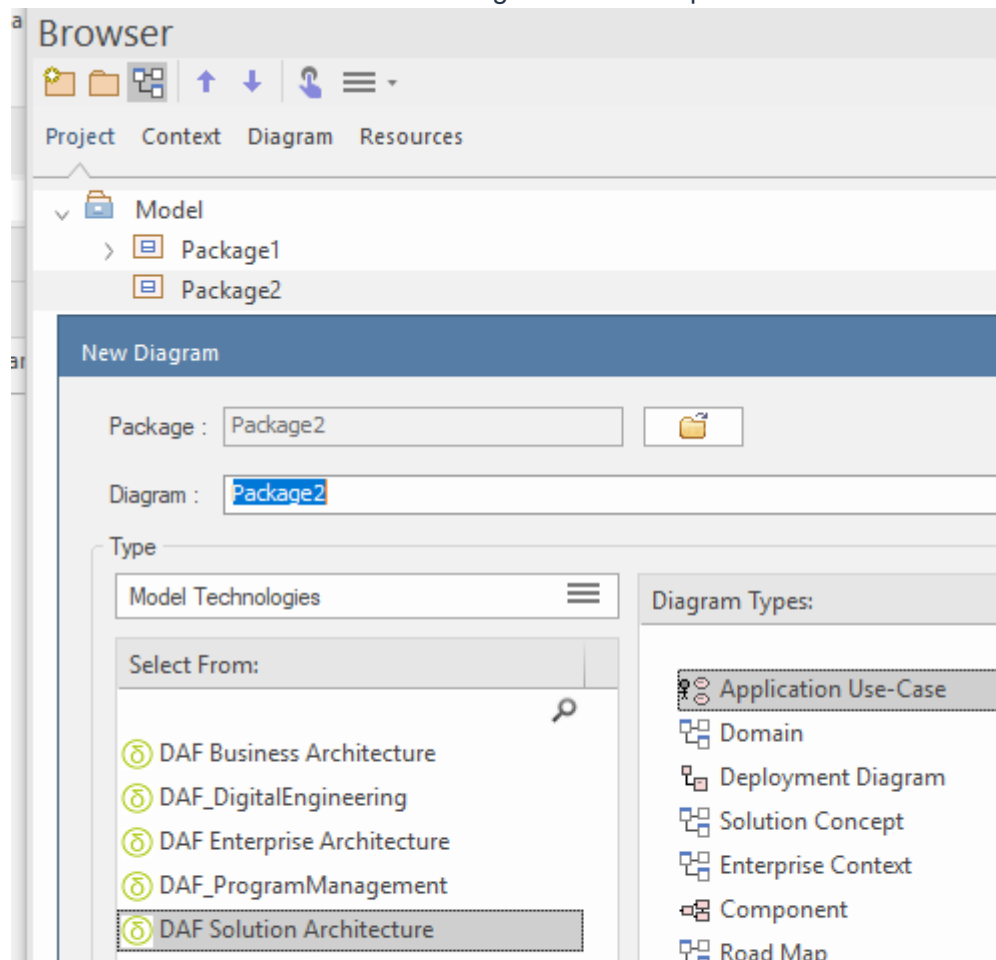


5. In the browser, open the tab **Resources** and navigate to **MDG technologies > Model Technologies**, right click and select **Import Technology**.
6. Locate and select the daf.xml file
7. Select **import to Model**
8. Click **OK** to load the MDG Technology into Sparx EA.
9. **Optional: Enable the DAF MDG Technology**
 - Go to **Specialize > Technologies > Manage Technologies**.
 - In the list of available MDG Technologies, find "DAF" and check the box to enable it.
 - Click **Close** to save your settings.
10. **Suggested:** set your perspective focus on DAF



11. Test a DAF diagram

- Create a new diagram
- Select one DAF Diagram from the options



5 Concepts

5.1 From Reality to Abstraction

DAF is a metamodel (M2) that describes the elements of a Model (M1) that is, in turn, an abstraction of a real system (M0).

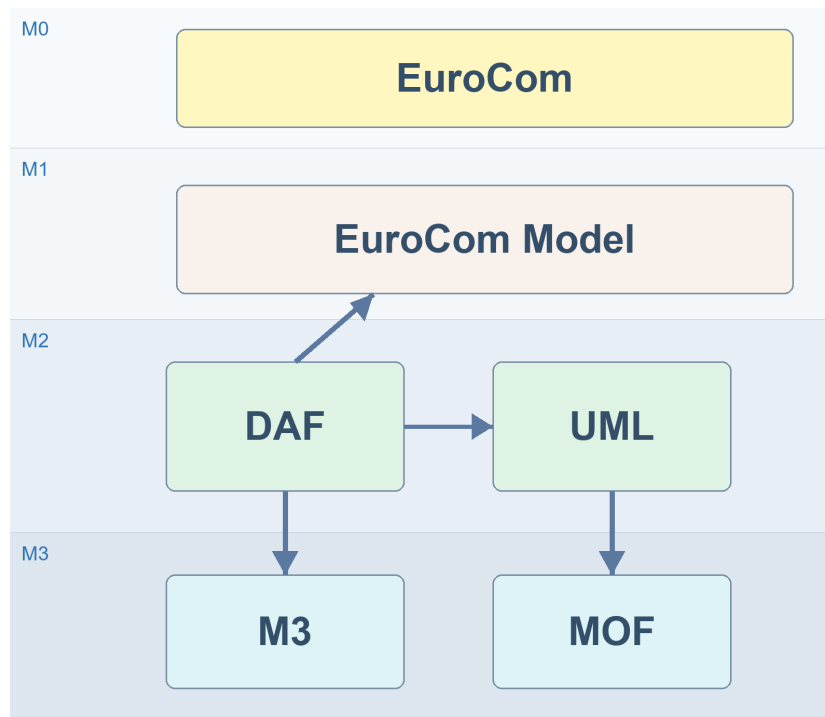


Figure 2: The hierarchy in this diagram follows the MDA principle of abstraction, where each level (M0 to M3) adds structure and abstraction to the level below, ensuring a consistent and scalable modeling approach.

- **M0: EuroCom:** The M0 layer represents the real-world organization or domain of interest, here as "EuroCom." This is the concrete instance of the domain that the models aim to represent, analyze, and improve. It consists of actual data and entities in the operational environment.
- **M1: EuroCom Model:** The M1 layer contains the model of the M0 domain. In this case, the "EuroCom Model" represents an abstraction of the EuroCom organization, describing its structure, behavior, or other aspects at a higher abstraction level. The M1 model acts as a blueprint that specifies how the M0 system is represented.
- **M2: DAF and UML:** The M2 layer defines the metamodels that describe the structure and semantics of the models in the M1 layer. The arrow from DAF to UML indicates influence, highlighting how DAF integrates concepts from UML.
- **M3: M3 and MOF:** At this level, the fundamental rules and concepts for defining modeling languages are established. The M3 layer represents the meta-metamodels that describe the foundation for defining metamodels like UML or DAF. M3 is a custom meta-metamodel that can describe DAF or other metamodels. M3 is used instead of MOF (Meta-Object Facility): A standard meta-metamodel defined by the OMG to provide a formal foundation for modeling languages like UML.

5.2 From the Model to a View

There is a formal way of separating the idea of "one model" from the way we look at it, develop it, and communicate it.

Metamodel	Model	Layers	Viewpoint	View
A model documenting the elements used in a specific area of interest and their relationships.	A set of entities that describes the area of interest and its relationships.	A level of abstraction within the considered model. There are three layers in the DAF model.	A way of looking at the underlying model for a particular type of observer. There are 12 viewpoints in the DAF model.	What is seen from a viewpoint. Diagrams, matrices, and catalogs are types of views.

The Digital Engineering approach aims to enable the separation of concerns across different models, metamodels and viewpoints, allowing for a more flexible, maintainable and understandable representation of the system, making it possible for different stakeholders to focus on the aspects of the system that are relevant to them.

A "**metamodel**" is a model ... of a model, which defines the structure and constraints of the models that conform to it. It describes the concepts, properties, and relationships that a model can contain, as well as the constraints and rules that apply to them. Metamodels are often expressed in formal languages, such as MOF or eMOF.

The concept of a "**model**" refers to a representation of a system or part of it in a specific format that adheres to the metamodel (e.g. UML or ArchiMate). These models can be used to define the architecture of a system. The ISO/IEC/IEEE 42010 standard defines "*architecture*" as "*the fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles guiding its design and evolution.*"

A "**Model Layer**" is a set of models that conform to a specific metamodel and are used to represent a specific aspect of the system being modeled. For example, one model layer could be used to represent the logical architecture of the system, while another could be used to represent the physical deployment of the system.

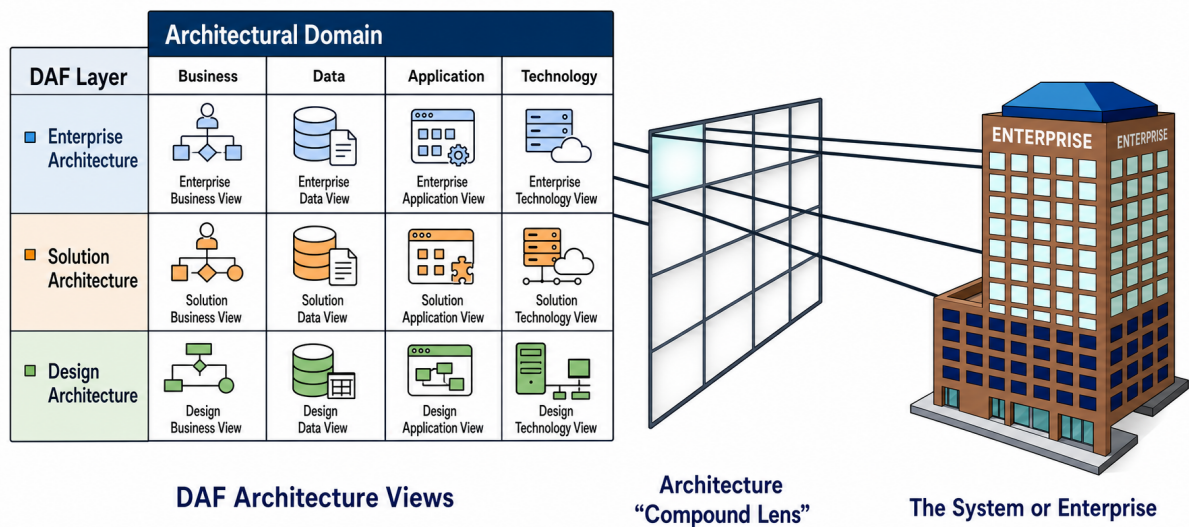


Figure 3: Relation between IT view and Enterprise

A **"Viewpoint"** is a set of views that have a common purpose or perspective. It can be related to specific domain, such as security or performance, and defines a specific set of views and model elements to be used to capture the relevant information from the system.

A **"View"** is a perspective or a specific part of a model that is relevant to a particular audience, such as stakeholders or developers. Each view provides a specific representation of the system, highlighting different aspects of the system and omitting the less important details.

5.4 Example Model Instance

Each element of the Metamodel can be instantiated into a concrete model's element.

The following diagram shows an example of a DAF Metamodel instance

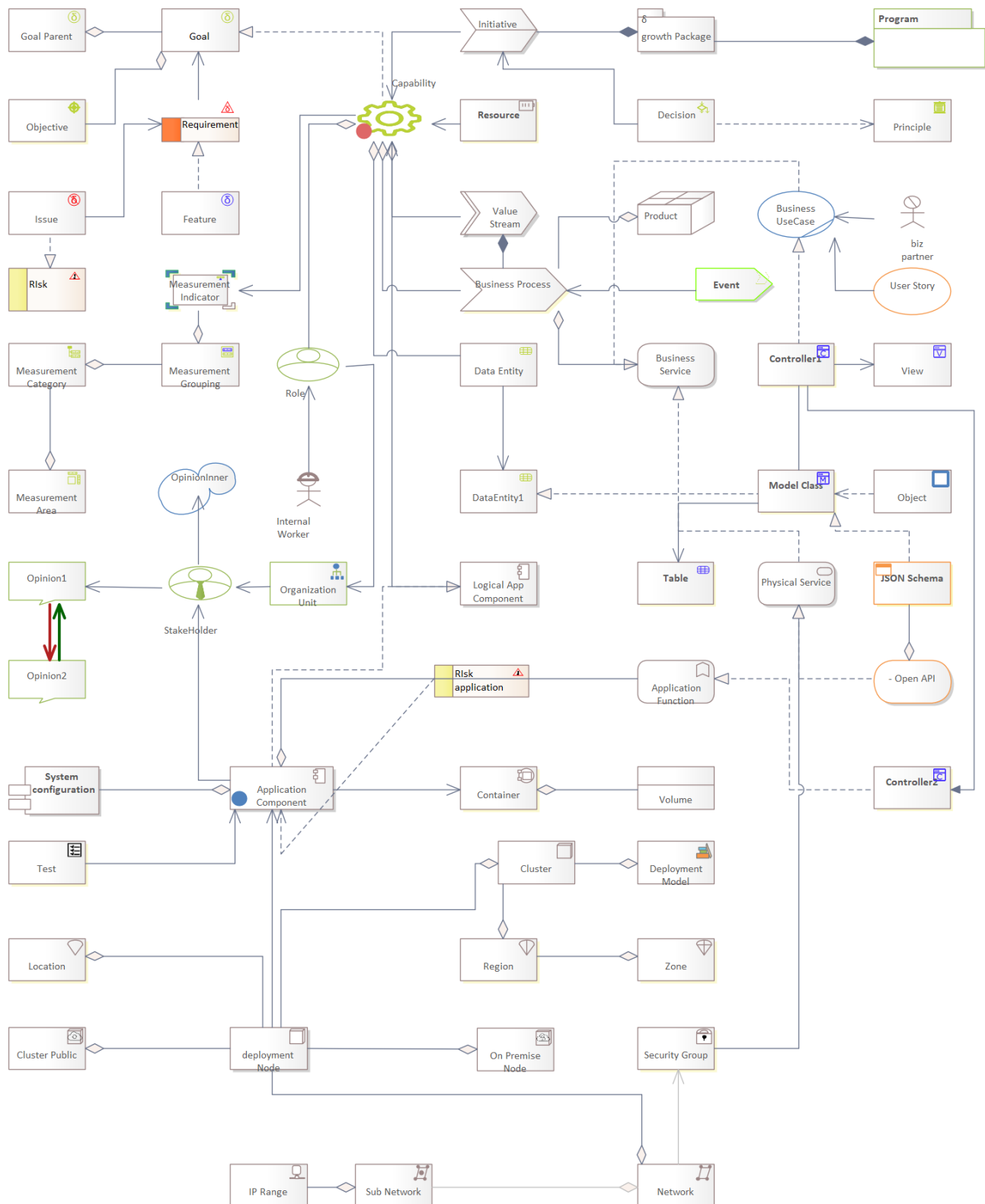


Figure 6: DAF 6.3 Example Instance

DAF supports the Sparx EA quick linker that helps modelers to create only valid connections. Models can be also validated using the Checker (see below)



5.5 Layers

DAF Layers

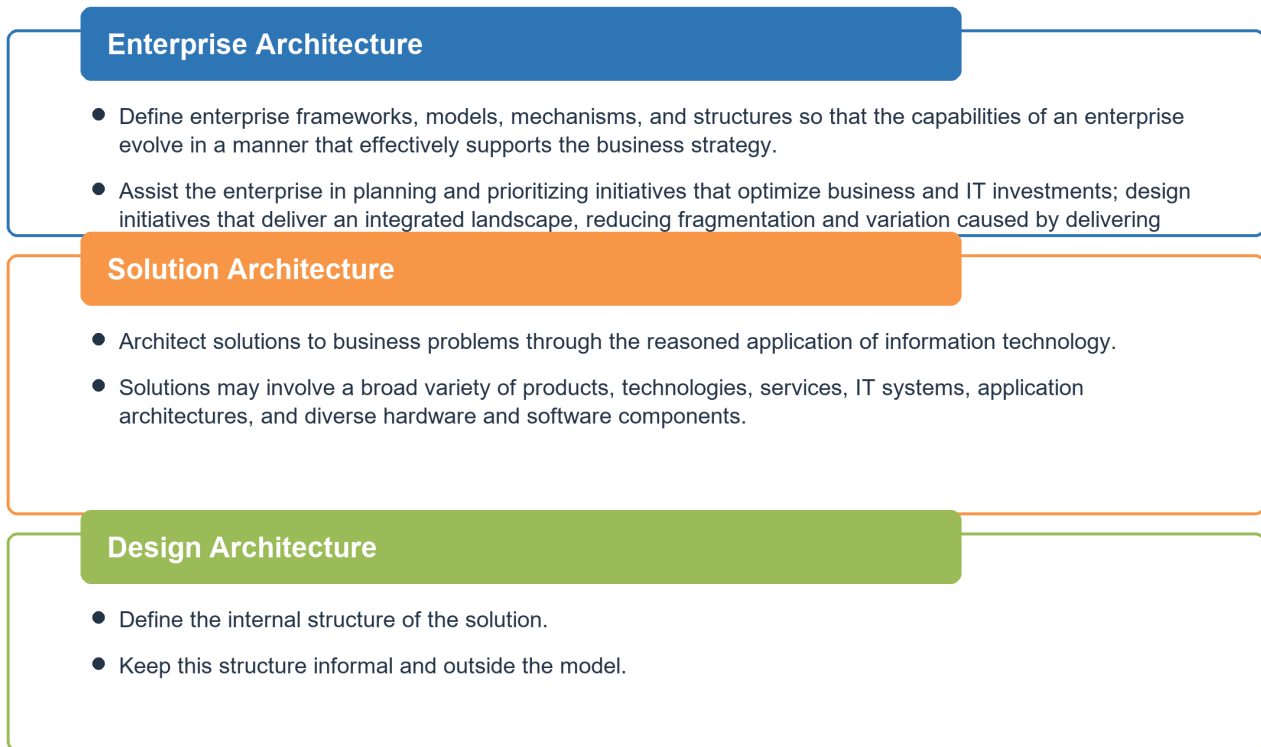


Figure 7: DAF Layers

DAF is divided into three levels of abstraction: Enterprise, Solution, and Design Architecture. Each of these levels is further broken into specific domains (see below).

5.5.1 Enterprise Architecture

- This Layer covers logical considerations. Viewing the system in terms of roles participating in processes by using resources: Who, How, and What. It corresponds to the Computational Independent Model (CIM).
- It defines enterprise frameworks, models, mechanisms, and structures so that capabilities of an enterprise evolve in a manner that effectively supports the business strategy
- Assisting the enterprise to plan and prioritize initiatives that optimize business & IT investments; designing initiatives that deliver an integrated landscape to reduces the fragmentation and variation that is the natural result of delivering solutions in an autonomous manner.
- The Enterprise Architecture level focuses on the logical considerations of the system, such as the roles, processes, and resources involved. This abstraction level helps the enterprise plan and prioritize initiatives that optimize business and IT investments, and design initiatives that deliver an integrated landscape to reduce fragmentation and variation.

5.5.2 Solution Architecture

- The Solution Architecture layer connects to the previous and translates the logical model into an operational one by describing the elements of the system that supports the business and assigns processes to them.
- It aims as definition of solutions to business problems through the reasoned application of Information Technology.

- They may involve the application and integration of a broad variety of products, technologies and services, various IT systems and applications architectures, and diverse hardware and software components.

5.5.3 Design Architecture

- This Layer designs the components of the solution, taking in account the different supporting technologies.
- The definition of the internal structure of the solution.
- Generation of Code

5.6 Domains

The abstraction layers are further organized into domains. A domain represents a **vertical organization** of concepts, incorporating related concepts at different levels of abstraction. These domains serve to organize and structure the various components and elements of the architecture in a logical and meaningful way.

The foundational domains in DAF are:

- Business
- Data
- Application
- Technology

Business Architecture	Data Architecture	Application Architecture	Technology Architecture
<i>A representation of holistic, multi-dimensional business views of: capabilities, end-to-end value delivery, information, and organizational structure; and the relationships among these business views and strategies, products, policies, initiatives, and stakeholders.</i> TOGAF	<i>A description of the structure and interaction of the enterprise's major types and sources of data, logical data assets, physical data assets, and data management resources.</i> TOGAF	<i>A description of the structure and interaction of the applications as groups of capabilities that provide key business functions and manage the data assets.</i> <u>TOGAF</u>	1. A technology building block. A generic infrastructure technology that supports and enables application or data components (directly or indirectly) by providing technology services. 2. An encapsulation of technology infrastructure that represents a class of technology product or specific technology product.

Figure 5: Foundational DAF domains, inspired by TOGAF

5.6.1 Business

This domain covers the business-specific concepts, such as business processes, services, and functions that an organization offers to its customers. It focuses on understanding the business objectives and strategies, and how the IT solutions can support those goals.

5.6.2 Data

This domain covers information-related concepts, such as data models, data architecture, data governance, data security, and data management. It's focused on understanding the organization's data requirements, and how the IT solutions can support data management and access.

5.6.3 Application:

This domain covers software-related concepts, such as application architecture, application integration, and service-oriented architecture. This domain is focused on understanding the organization's application requirements, and how the IT solutions can support application deployment and management.

5.6.4 Technology

This domain covers technology-related concepts, such as infrastructure, network, security, and cloud. This domain is focused on understanding the organization's technology requirements, and how the IT solutions can support infrastructure and technology management.

Biz Architecture	Data Architecture	Application Architecture	Technology Architecture	Implementation Architecture	Requirement Architecture
Product	Data Entity	Logical App Component	Location	Program	Goal
Business Process	Model Class	Application Function	Node	Growth Package	Requirement
Business Service	Table	Application Component	Network	Initiative	Feature
Value Stream	Object	Physical Service	System Configuration	Decision	Risk
BusinessArea	Test	View		Principle	Issue
Capability		Controller		Stakeholder	Measurement Grouping
Business Use Case				Opinion	Measurement Area
Org Unit				Inner Opinion	Measurement Category
Role					Measurement Indicator
Objective					
Event					

Figure 6: Domains and concepts of DAF

5.6.5 Extension Domains

Additional Domains can be also used with DAF when required.
Those include Strategy, Implementation and Requirements.

5.7 Viewpoints and Views

The intersection of a Layer and Domain creates a Viewpoint, addressing specific concerns about a system of interest. A Viewpoint is a way of looking at a system from a specific perspective and addressing specific concerns related to that system. In the context of DAF, a **Viewpoint** is the intersection of a Layer and a Domain.

Each Layer, such as Enterprise, Solution, and Design Architecture, provides a different level of abstraction and focus, while each Domain, such as Business, Data, Application, and Technology, represents a specific area of concern. By intersecting these two concepts, a Viewpoint is created that addresses specific concerns of the system-of-interest from the perspective of that Layer and Domain. For example, a Business Viewpoint within the Enterprise Layer would focus on the logical considerations of how the system supports the business strategy, while a Data Viewpoint within the Design Layer would focus on the design of the data components of the solution.

The use of Viewpoints allows for a more targeted and granular approach to analyzing and designing a system, as different stakeholders can look at the same system from different perspectives.

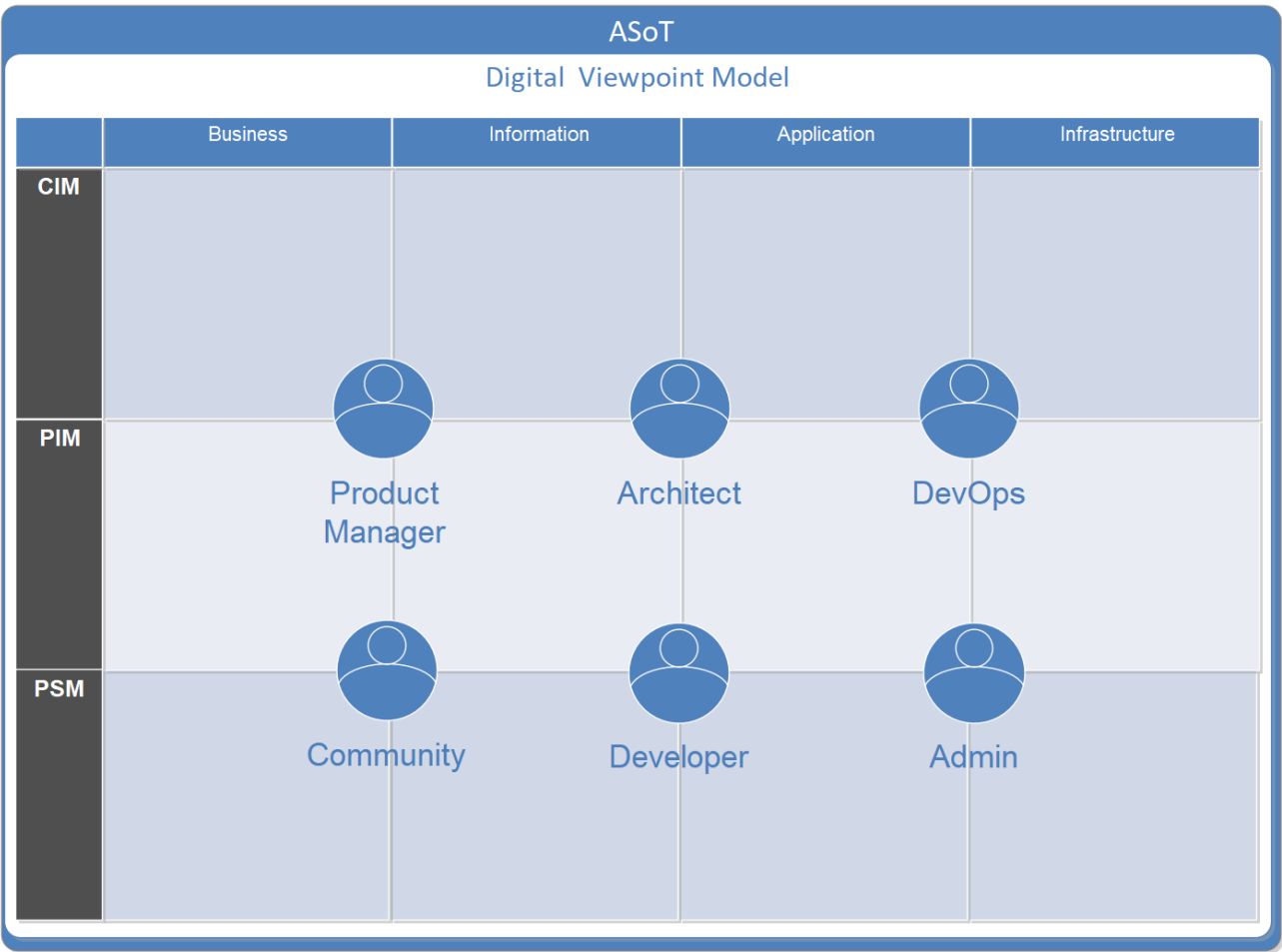


Figure 7: Digital Viewpoint Model

DAF features 12 basic viewpoints

Figure 8: example of viewpoint and their stakeholders

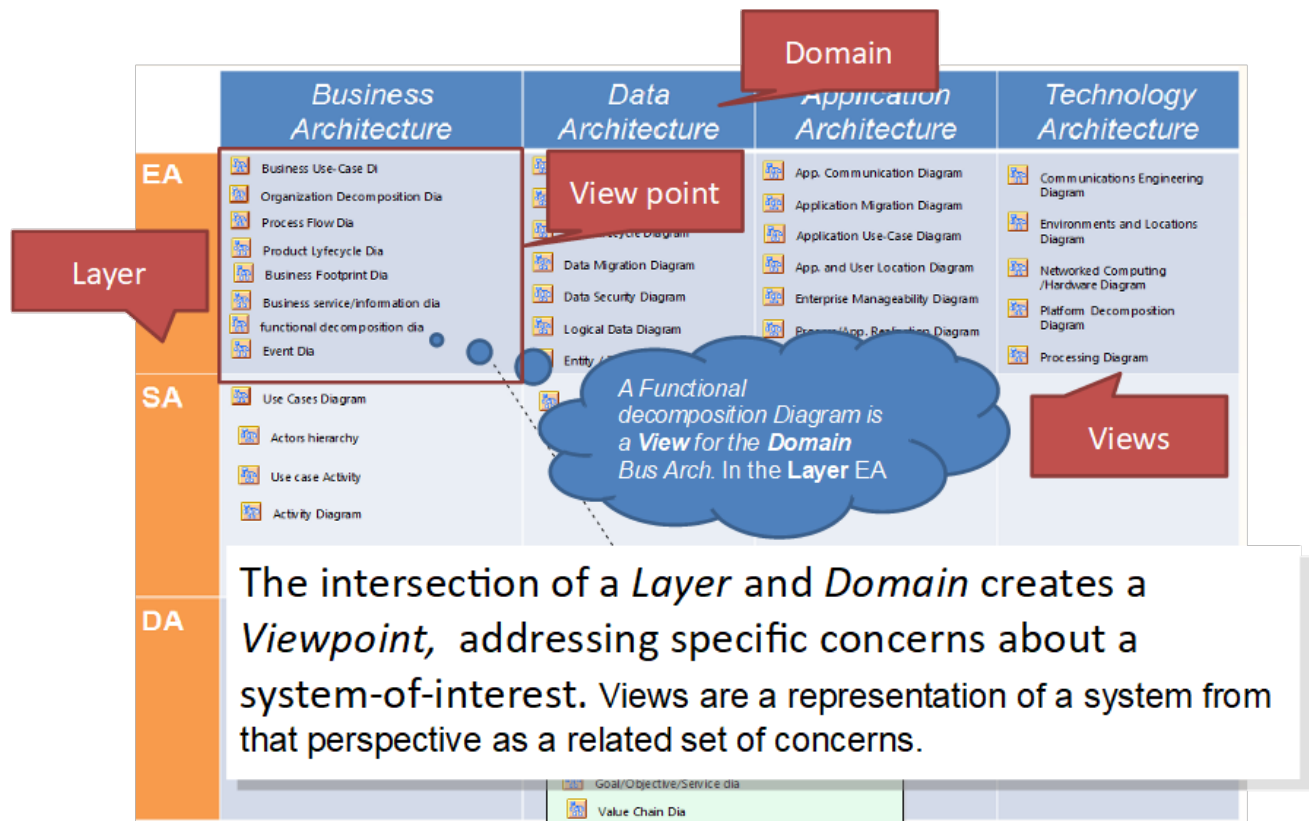


Figure 8: DAF Viewpoints

Views are representations of a system from that perspective as a related set of concerns.

5.7.1 DAF Concepts Organized in Viewpoints

Each of the concepts of the Metamodel belong to a specific Viewpoint. As such, it's easier to assign responsibility to certain roles.



Figure 9: DAF Concepts Organized in the Viewpoints

5.7.2 Views and work products

DAF supports the SPEM concept of Deliverable.

A **Deliverable** is created when a **role** performs an activity within a process that produces one or more views. Each view is built from the DAF entities stored in the repository catalog. Multiple views together form a complete deliverable.

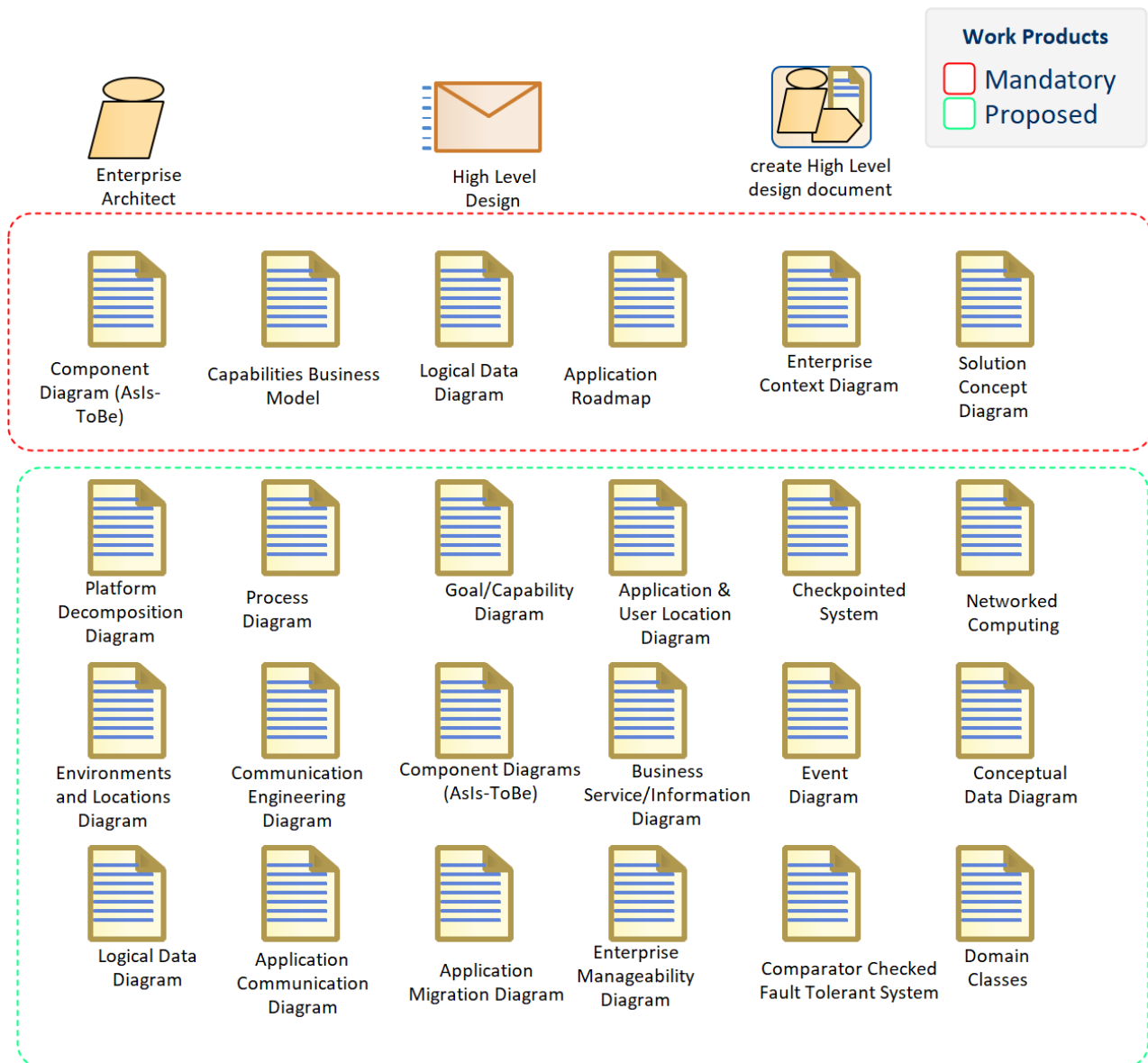


Figure 10: SPEM View with the Work Products That Compose the High-Level Design Deliverable

In the example above, the Enterprise Architect performs the activity *Create High-Level Design Document*. The result is the **High-Level Design** deliverable. This includes mandatory work products, such as the *Component Diagram (As-Is/To-Be)* and the *Capability Business Model*. Depending on the scope, it may also include optional work products, like the *Platform Decomposition Diagram* or the *Process Diagram*.

By its nature, a deliverable is like a 2D photo: it presents a static view that does not necessarily expose all implications or dependencies. However, the underlying model contains the full traceability, which can be retrieved to perform impact analysis and uncover relationships not immediately visible in the deliverable itself.

5.8 Relationships

Name	Meaning	Notation
Association	<i>"has"</i>	
Aggregation	<i>"shared with"</i>	
Composition	<i>"owned-by"</i>	
Dependency	<i>"uses"</i>	
Generalization	<i>"is-kind-of"</i>	
Realization	<i>"implements"</i>	

Figure 11: Basic UML relationships

DAF relationships are extending fundamental UML relationships:

- **Association:** A relationship that models a bi-directional semantic connection among instances
- **Aggregation:** A special kind of association that models a "whole-part" as shared relationship
- **Composition:** non-shareable, non-transferable ownership
- **Dependency:** A relationship that signifies that a single or a set of model elements requires other model elements
- **Generalization:** A taxonomic relationship between a more general element and a more specific element
- **Realization:** A specialized abstraction relationship between two sets of model elements, one representing a specification (the supplier) and the other represents an implementation of the latter (the client)

In the DAF metamodel, inheritance uses Generalizations, transitions of abstraction layers are represented with stereotyped Realization, strong connections between elements of the same abstraction level with stereotyped **Associations**, weak relationships with Dependencies; strong hierarchical relationships with **Compositions**, weak ones with **Aggregations**.

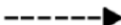
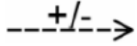
Name	Meaning	Notation
Triggering	Describes a temporal or causal relationship between elements.	
Flow	Transfer from one element to another.	
Assignment	Expresses the allocation of responsibility, performance of behavior, or execution.	
Serving	Models that an element provides its functionality to another element.	
Access	Models the ability of behavior and active structure elements to observe or act upon passive structure elements.	 
Influence	Models that an element affects the implementation or achievement of some motivation element.	

Figure 12: Additional ArchiMate relationships

5.8.1 Relationship Strategy

The DAF Metamodel is designed to support the concept of **encapsulation**.

Encapsulation is a key concept in modeling that refers to the practice of enclosing concept and behaviors within a single entity, so as to protect its internal state from external interference. This helps to maintain the integrity of the model by controlling the flow of information and avoiding circular references. In object-oriented modeling, encapsulation allows objects to interact with one another without being dependent on their implementation, creating a level of abstraction that makes the system more flexible and maintainable.

Circular references in modeling refer to the situation where two or more models or elements within a model refer to each other in a circular manner. This creates a situation where one element depends on another, which in turn depends on the first, leading to a never-ending chain of inter-dependencies. Circular references are often considered problematic in modeling because they can lead to ambiguities and inconsistencies in the models, making it difficult to analyze and understand the relationships between elements. They can also cause problems with automated analysis and interpretation tools that rely on the models.

Instead of circular references, **derived relationships** are used.

A derived relationship refers to a relationship that is obtained or computed from other relationships or information in the model. It is not directly specified in the model but is instead derived from existing elements and their properties and relationships. Derived relationships are often used to simplify a model and make it more manageable by abstracting complex relationships and showing only the essential relationships between elements. These relationships may also be used to generate additional information about the system being modeled or to support certain modeling activities such as impact analysis, traceability, and consistency checks.

6 Detailed Documentation

The following is the complete documentation of the DAF metamodel

6.1 DAF Concepts Overview

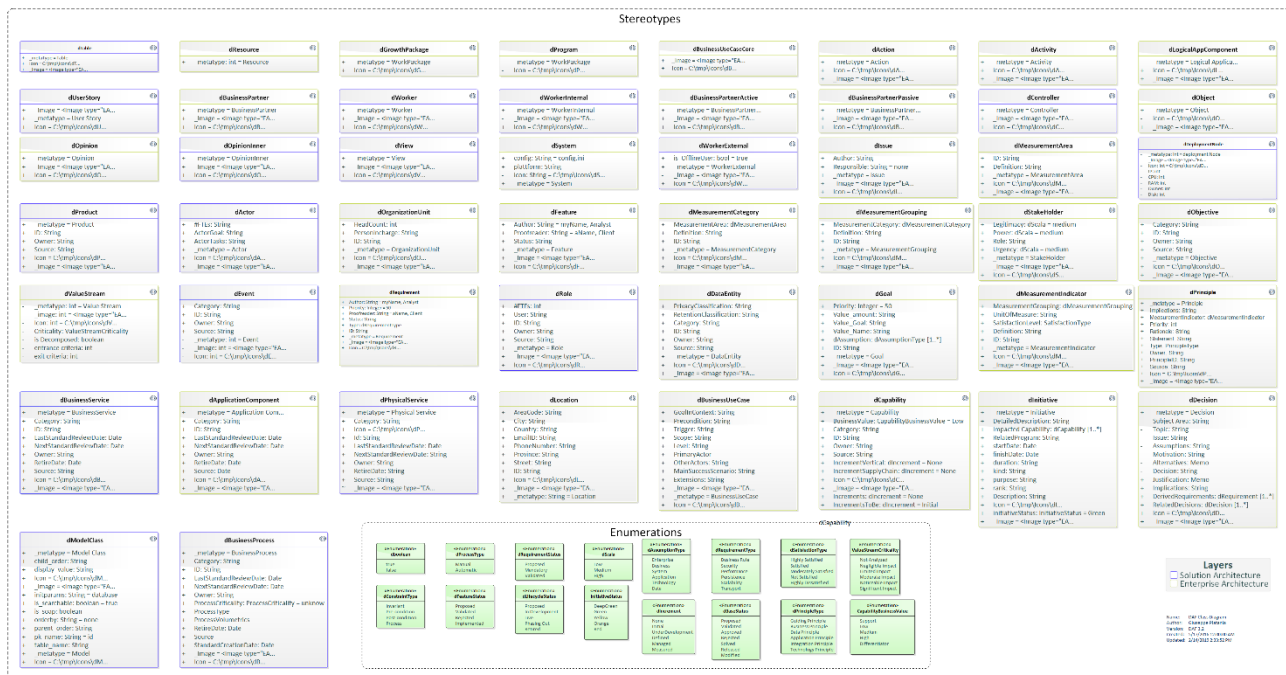


Figure 13: DAF Concepts and Properties

DAF version 6.3 supports ~71 concepts (implemented as UML Stereotypes), each of those has several properties (expressed as UML Tagged Values)

Domain	Technical Name	Human Name	Documentation
Application	dTest	Test	Represents a mechanism for verifying or validating the functionality, behavior, or compliance of an application component. Tests can include automated or manual procedures designed to ensure alignment with requirements, detect defects, and confirm expected outcomes.
Application	dModelClass	Model Class	Represent a class of the domain model, that can be converted into a Persistent domain class, must inherit from Node. It is the application's dynamic data structure
Application	dLogicalAppComponent	Logical App Component	Encapsulates logical software components independent of implementation or technology.
Application	dApplicationFunction	Application Function	Encapsulates functions supported by applications.
Application	dController	Controller	Encapsulates application behavior and flow logic.
Application	dValue	Value	An attribute of a Node value type used in DAF. These values are persistent
Application	dAppComponent	Application Component	An application component is a self-contained unit. As such, it is independently deployable, re-usable, and replaceable. An application component performs one or more application functions. It encapsulates its behavior and data, exposes services, and makes them available through interfaces. Cooperating application components are connected via application collaborations. An application component may be assigned to one or more application functions. An application component has one or more application interfaces, which expose its functionality. Application interfaces of other application components may serve an application component. The name of an application component should preferably be a noun.

Domain	Technical Name	Human Name	Documentation
Application	dView	View	A View used in DAF. Controller, Views and Associations define the application flow
Application	dPhysicalService	Physical Service	a technical implementation of a service that runs on physical or virtual infrastructure, providing specific functionalities, usually in the form of software applications, middleware, databases, or platforms. It is concerned with the deployment, operation, and management of the technology stack, ensuring that the service is available, performant, and reliable. Examples include web servers, APIs, databases, and network services.
Application	dValueRef	Value Ref.	A Node value type used in DAF that reference another Node Value, similar to a foreign Key. These values are persistent
Business	dRole	Role	The usual or expected function of an actor, or the part somebody or something plays in a particular action or event. An actor may have a number of roles.
Business	dActivity	Activity	Represents business or system activities. A function or work carried out by the business user.
Business	dActor	Actor	Represents an entity (such as a person, system, or organization) that interacts with the system to achieve a specific goal. An actor typically performs actions within business Use cases and can be internal (e.g., employees) or external (e.g., customers or partner organizations), active and passive.
Business	dBusinessUseCase	Business Use Case	Encapsulates business processes producing value for an enterprise.
Business	dBusinessService	Business Service	A system configuration. Settings are modeled as attributes with default values

Domain	Technical Name	Human Name	Documentation
Business	dAction	Action	A specialized activity.
Business	dEvent	Event	A behavioral situation that causes a change in the state of an element (typically a Process)
Business	dBusinessProcess	Business Process	A sequence of activities that produce a specific service or product supporting a particular business capability.
Data	dTable	Table	Represents physical tables in a database schema.
Data	dObject	Object	an Object is an instance of a class in a particular moment in runtime that can have its own state and data values.
Implementation	dDecision	Decision	Represents decisions and their impact on architecture or processes.
Implementation	dUserStory	User Story	Captures user-centric requirements expressed as stories.
Implementation	dInitiative	Initiative	An activity carried out by a Business Unit or organization to achieve a particular goal or improve the Maturity of a certain capability.
Implementation	dGrowthPackage	Growth Package	A set of actions identified to achieve one or more objectives for the business. A work package can be a part of an Initiative, a complete project, or a part of a Program. A Growth pack is a collection of Initiatives architected for consistency and aimed to be executed in the same period.
Implementation	dProgram	Program	A group of Growth Packages that achieve a complex outcome for the organization

Domain	Technical Name	Human Name	Documentation
Requirement	dRisk	Risk	Identifies and assesses potential issue impacting other elements. A risk is a potential future Issue.
Requirement	dFeature	Feature	A Feature is a fact that set his related Requirement to be true
Requirement	dIssue	Issue	A concrete problem that is affecting a business. Also, a fact that sets a Requirement to be false. (see also Risk)
Requirement	dRequirement	Requirement	A business guideline about the Enterprise or the project. A requirement is formulated in a SMART fashion and uses Moscow verbs (Must Have, Should Have, Could Have, and Would Like to Have.). Requirements can be decomposed. • Must Have: These are the requirements that are considered essential for the success of the project and must be met for the system to function. They are often referred to as "hard" or "mandatory" requirements and are typically expressed in Boolean format as "true" or "1". • Should Have: These are the requirements that are considered important, but not essential. They are typically expressed in Boolean format as "maybe" or "0/1" and are often used to prioritize the development of the system. • Could Have: These are the requirements that are considered desirable but not essential. They are typically expressed in Boolean format as "false" or "0". They are used to identify potential enhancements to the system that can be added at a later date. • Would Like to Have: These are the requirements that are considered less important and are often used to identify potential future enhancements to the system. In addition, the requirement should be SMART, which stands for Specific, Measurable, Achievable, Relevant, and Time-bound. SMART requirements are clear, concise, and easy to understand, and they provide a clear way to measure the success of the project.
Requirement	dGoal	Goal	A measurable scope that the organization wants to achieve. Can be hierarchically decomposed.
Strategy	dMeasurementIndicator	Measurement Indicator	The specific measure captured for an organization or Business Unit which can be measured and quantifiable. A measurement is a Key Performance indicator
Strategy	dMeasurementArea	Measurement Area	Represents areas of organizational or business performance measurement. The organization or

Domain	Technical Name	Human Name	Documentation
Strategy	dStakeholder	Stakeholder	Business units' performance objective is captured as measurement area.
			Represents any party or force influencing the domain of interest. Any person or force that has an interest in the considered domain. This is typically indicated by name, can be connected with a Role in the organization. A stakeholder has open and Inner opinions that are relevant for the subject matter.
	dCapability	Capability	Represents an organizational capability to achieve business outcomes. Can be currently available in an Organization Unit (or) A business-focused outcome that is delivered by the completion of one or more work packages (by increasing the capability Maturity).
	dProduct	Product	Encapsulates outputs produced by business processes.
	dPrinciple	Principle	Defines qualitative statements guiding architecture intent.
	dOrganizationUnit	Org Unit	Defines organizational units responsible for roles, can be decomposed
	dMeasurementCategory	Measurement Category	Categorizes measurements within a measurement area.
	dOpinionInner	Inner Opinion	Captures implicit stakeholder opinions related to a topic. A Stakeholder's opinion that is relevant for the topic but is not formulated explicitly.
	dSkill	Skill	an ability that some role possesses with various degree
	dOpinion	Opinion	A view or judgment formed about a Topic. That is formulated in an explicit fashion. Captures explicit stakeholder judgments on a topic.
Strategy	dResource	Resource	a type of Asset supporting a Business Capability

Domain	Technical Name	Human Name	Documentation
Strategy	dObjective	Objective	A time-bounded milestone for an organization used to demonstrate progress towards a goal. The Objective is a non material achievement, other than a Goal, the objective has no specific value to be measured
Strategy	dValueStream	Value Stream	A representation of an end-to end collection of value-adding Business Processes that create an overall result for a customer, stakeholder or end user
Strategy	dLocation	Location	A place where business activity takes place and can be hierarchically decomposed.
Strategy	dMeasurementGrouping	Measurement Grouping	a group of Measurement Indicator that can be aggregated into Categories
Technology	dJSON_Schema	JSON Schema	The Schema Object allows the definition of input and output data types. These types can be objects, but also primitives and arrays. This object is based on the JSON Schema Specification Draft 4 and uses a predefined subset of it. On top of this subset, there are extensions provided by this specification to allow for more complete documentation
Technology	dDeploymentModel	Deployment Model	The root class for the description of deployments to roll out an application, service, or container to a set of resources. similar to ArchiMate's Technology Layer deployment views.
Technology	dNetwork	Network	The main communication medium between services and deployments, mapped to ArchiMate Communication Networks, facilitating inter-service communication.
Technology	dJSON_SchemaSubSet	JSON Schema Sub Set	Represents subsets of JSON schema structures.
Technology	dAPIResponse	API Response	Represents responses returned by API operations.

Domain	Technical Name	Human Name	Documentation
Technology	dJSON_Element	JSON Element	Represents elements or nodes in JSON schemas.
Technology	dJSON_Type	JSON Type	Represents data type definitions for structured information.
Technology	dCluster	Cluster	Represents a set of nodes working in the same Region, similar to ArchiMate's Node concept within a server infrastructure.
Technology	dDeploymentNode	Node	Represents a physical machine or virtual server that hosts services, applications, or containers, similar to ArchiMate's Node concept in the Technology Layer.
Technology	dJSON_Datatype	JSON Data Type	Represents a data type used within JSON structures, defining the format and constraints of data elements in a schema. JSON Data Types include standard types such as string, number, object, array, boolean, and null, ensuring consistency and validation in data exchange between systems.
Technology	dVolume	Volume	Refers to the storage assigned to containers, ensuring persistence, aligning with the Data Object in ArchiMate's Application Layer.
Technology	dAPI	Open API	OpenAPI Specification is an open-source format and initiative for designing and creating machine-readable interface files that are utilized in producing, describing, consuming, and visualizing RESTful APIs and web services.
Technology	dContainer	Container	Encapsulates application components and services, mirroring ArchiMate's concept of a Node, where an application runs.
Technology	dAPIParameter	API Parameter	Defines parameters in API operations.

Domain	Technical Name	Human Name	Documentation
Technology	dAPIOperation	API Operation	Defines operations exposed by an API for interaction with external systems. Also called endpoints.
Technology	dZone	Zone	Defines macro divisions containing regions mapped to ArchiMate's Infrastructure Service.
Technology	dJSON_Attribute	JSON Attribute	Defines attributes in JSON data structures.
Technology	dIPRange	IP Range	Defines a range of IP addresses (CIDR) used within a network. This aligns with ArchiMate's concept of Infrastructure Service, providing connectivity.
Technology	dSecurityGroup	Security Group	Controls and manages access and permissions between services, in line with TOGAF's Security Architecture for protecting digital assets.
Technology	dOnPremise	On Premise Node	an organization of Business interests
Technology	dDataEntity	Data Entity	An encapsulation of data that is recognized by a business domain expert as a thing. Logical data entities can be tied to applications, repositories, and services and may be structured according to implementation considerations.
Technology	dSystem	System Configuration	A system configuration. Settings are modeled as attributes with default values; each attribute of the System Config becomes a line in the configuration file. e.g. SystemConfig.path = c:/users/app
Technology	dSubNetwork	Sub Network	A subset of a network, used to organize and isolate portions of the network. Equivalent to an ArchiMate Node, responsible for networking resources.
Technology	dPublicCluster	Public Cluster	A public cloud-based cluster, following the ArchiMate concept of Infrastructure Service provided by external cloud providers (Google, AWS, etc.).
Technology	dRegion	Region	A geographic area containing clusters and zones, aligned with ArchiMate's Location concept for

Domain	Technical Name	Human Name	Documentation
			representing deployment geography.

6.2 DAF Relationship Types

Technical name	Human Name	Destination	Source
dAccord	According Opinion	is in accord with other Opinion	is in accord with original Opinion
dActionKey	Action Key	target Controller	Source Controller
dActivityContainsAction	Activity contains Action	is contained in Activity	contains Action
dActorParticipatesInBizUseCase	Actor Participates In Use Case	Partecipates in Use Case	is executed by Actor
dApplicationAgregatesFunction	aggregates Function	aggregates Function	part of Application
dApplicationComponentRealizesogicalComponent	Realizes Component	Realizes Logical component	is Realized by Component
dApplicationconfiguredbySystemConfiguration	Application configured by System Configuration	has configuration	configures Application
dApplicationConnectedWithApplication	Connects Application	connected From Application	Connected to Application
dApplicationExposesService	Application Exposes Ph. Service	is exposed by Application	Exposes Phy Service

Technical name	Human Name	Destination	Source
dApplicationHasOwner	Application has Owner	owns Application	has Owner
dApplicationIRunsInNode	Application is run By Node	Runs Application	is run By Node
dApplicationRisk	Application Risk	is related to Application	has Risk
dAreaAggregatesCategories	Area aggregates Categories	aggregates categories	has Measurement Area
dBizServIsDescribedByUseCase	Biz Serv Is Described By Use Case	Has service	described by Use Case
dBizServUsesDataEntity	Service Uses Data	uses Data	is used by service
dBusinessAreaHasCapability	has Capability	has Capability	is organized in Biz Area
dCapabilityIsMeasuredByKPI	is Measured By KPI	is Measured By Indicator	is measuring Capability
dCategoryAggregatesGroups	aggregates Groups	aggregates Groups	has Category
dContainerAggregatesVolume	Container Aggregates Volume	Aggregates Volume	is Aggregated in Container
dControllerControlsModel	Controls Model	Controls Model	is controlled by Controller
dControllerGovernsView	Governs View	Governs view	is Governed by Controller

Technical name	Human Name	Destination	Source
dControllerImplementsFunction	controller Implements function	Implements function	has Controller
dDataEntityInformsCapability	informs Capability	is informed by Data Entity	Informs Capability
dDecisionRefersToPrinciple	Decision refers To Principle	refers To Principle	is referenced by Decision
dDiscord	Discording opinion	is in discord with other Opinion	is in discord with Original Opinion
dEntityAssociatesEntity	Entity Associates Entity	Is connected to Source Entity	Is connected to Target Entity
dEntityisFunctionallyImplementedByTable	Entity is Functionally Implemented By Table	Implements Entity	is Functionally Implemented By Table
dEventTriggersProcess	Event Triggers Process	triggers Process	is triggered by Event
dFeatureIsRealizedByUseCase	Feature is realized by Use Case	realizes Feature	is realized by Use Case
dGoalAggregatesGoal	Goal Aggregates Goal	parent Goal	Child Goal
dGoalhasObjective	Goal has Milestone	Has Objective	is related to Goal

Technical name	Human Name	Destination	Source
dGoalisOperationalizedByCapability	goal is Operationalized By Capability	support Goal	is Operationalized By Capability
dGroupAggregatesKPI	Group aggregates KPI	aggregates KPI	part of a Grouping
dInitiativeIncreasesMaturityOf	Initiative Increases Maturity Of Capability	Increases Maturity Of Capability	is matured by Initiative
dInitiativeisMakingUseOfDecision	Initiative is Making Use Of Decision	is used in Initiative	Makes Use Of Decision
dIssueStopsRealizationOfRequirement	Issue Stops Realization Of Req	stop Requirement	is stopped by Issue
dLocationHostsNode	Location Hosts Node	Hosts Node	Is hosted in Location
dLogAppCompExposesBizService	exposes Biz Service	application runs service	service run by application
dLogicalApplicationComponentSupportsCapability	App Supports Capability	supported by Application	supports Capability
dModelClassRealizesEntity	Model Class Realizes Entity	Realizes Entity	is realized by Class
dNetworkContainsNode	Contains Node	Contains Node	is contained in a Network

Technical name	Human Name	Destination	Source
dNetworkHasSecurityGroup	Network Has Security Group	Has Security Group	is associated to Network
dObjectInstanceofController	Instance of Controller	Has an Instance	Instance of Controller
dObjectInstanceofModel	Instance of Model	Has an Instance	Instance of Model
dObjectInstanceofView	Instance of View	Has an Instance	Instance of View
dOrgUnitHasStakeholders	Org Unit has Stakeholders	has Stakeholder	is concerned with Organization
dPackageIsDeliveredByInitiative	Package is delivered by Initiative	is delivered by Initiative	is part of a Package
dPhysicalServiceImplementsBusinessService	Physical Serv. Implements Biz Serv.	Implement Biz Service	is Implemented by physical service
dProcessEnsureCorrectOperationOfCapability	Ensure the correct Operation of	is operated by Process	Ensure the correct Operation of Capability
dProcessFlowsToProcess	Process Flows to Process	Sequence Flow To	Sequence Flow From
dProcessOrchestratesService	Process Orchestrates Service	Has Service	part of Process

Technical name	Human Name	Destination	Source
dProductIncludesProcess	Product Includes Process	Includes Process	Is part of Product
dProgramIsOrganizedInPackages	Program is organized In Packages	is organized In Packages	part of a Program
dRelatedService	Child Process	Target Service	Source Service
dRequirementContainsRequirement	Requirement Contains Req	is contained by Requirement	Contains Requirement
dRequirementIsRelatedToGoal	req related to Goal	specifies Goal	is specified by Requirement
dRequirementIsSatisfiedByFeature	is Satisfied By	realizes Requirement	is realized by Feature
dResourceImpactedByRisk	Resource impacted by Risk	impact Resouce	is impacted by Risk
dResourceSupportsCapability	Resource supports Capability	Supports Capability	is supported By Resource
dRiskIsRelatedToRisk	Risk is related to Risk	target Risk	Source Risk
dRiskRealizedByIssue	realized by Issue	is realizing Risk	is realized by Issue
dRoleConsumesService	Role Consumes Service	Consumes Service	is consumed by Role

Technical name	Human Name	Destination	Source
dRoleExecutesCapability	role Executes Capability	is executed by Role	executes Capability
dRoleHasSkill	Role has skill	has SKill	known by Role
dRoleIsAssociatedToOrganization	Role is Associated To Organization	works in Organization	has Role
dRoleIsPerformedByActor	Role is Performed By Actor	has Role	is executed by Actor
dStakeholderHasInnerOpinion	Stakeholde Has Inner Opinion	Has Inner Opinion	is thought by Stakeholder
dStakeholderHasOpinion	Stakeholder Has Opinion	Has Opinion	is expressed by Stakeholder
dTableProvidesPersistenceForModelClass	Table Provides Persistence to Model	is Saved in a Table	Provides Persistence For Model
dTest_ApplicationComponent	Test validates Component	Test validates Component	is validated by test
dUseCaseContainsActivities	Use Case Contains Activities	has Activitiy	part of UseCase
dUseCaseHasStory	Use Case Has Story	Is associated to Use Case	Can be told as User Story

Technical name	Human Name	Destination	Source
dUseCaseisImplementedByController	Use Case is Implemented B y Controller	implements Use Case	Is implemented by Controller
dValueStreamIncludesProcesses	Value Stream Includes Processes	Includes Process	is part of Value Stream
TMF_isComposedByProcess	is Composed By Process	is Composed By Process	is Composed within Process
TMF_ProcessProducesEntity	Produces Entity	Process produces Entity	Entity is produced by process
TMF_ProcessUsesEntity	Process Uses Entity	Uses Entity	is used by Process
TMF_RelatedUseCase	Related Use Case	target related Use Case	related Use Case
TMF_ThemeComposesUserStory	Composes User-Story	Composes User-Story	is composed In Theme
TMF_UseCaseAssociatedWithTestCase	Associated With Test Case	Associated With Test Case	Associated With Use-Case
TMF_Use-CaseDependOnTool	Depends On Tool	Depends On Tool	Is depended of Use-Case
TMF_Use-CaseDependsOnPolicy	Depends On Policy	Depends On Policy	is dependent of Use-Case
TMF_Use-CaseDependsOnUser-Story	Depends On User-Story	Depends On Use-Case	Depends On User-Story

Technical name	Human Name	Destination	Source
TMF_UseCaseRelatedProcess	Related Process	related to Process	related to Use Case
TMF_ValueStreamAggregatesValueStage	Aggregates Value Stage	Aggregates Value Stage	is aggregated by Value Stream
TMF_VerticalAggregatesProcess	Vertical Aggregates Process	Aggregates Process	Is aggregated in Verticals
TMF_VerticalComposedVertical	Vertical Composes Vertical	Child Vertical	Parent Vertical

6.2.1 DAF Relationships Quick link

Source → Relationship Name → target	Source → Relationship Name → target
dActivity -> dActivityContainsAction -> dAction	is contained in Activity
contains Action	dActor -> dActorParticipatesInBizUseCase -> dBusinessUseCase
Participates in Use Case	is executed by Actor
dActor -> dRoleIsPerformedByActor -> dRole	has Role
is executed by Actor	dAPI -> dAPIAggregatesSchema -> dJSON_Schema
Aggregates Schema	is aggregates in API
dAPI -> -> dPhysicalService	dApplicationComponent -> dApplicationComponentRealizesogicalComponent -> dLogicalAppComponent
Realizes Logical component	is Realized by Component
dApplicationComponent -> dApplicationconfiguredbySystemConfiguration -> dSystem	has configuration
configures Application	dApplicationComponent -> dApplicationExposesService -> dPhysicalService

Source → Relationship Name → target	Source → Relationship Name → target
is exposed by Application	Exposes Phy Service
dApplicationComponent -> dApplicationHasOwner -> dStakeholder	owns Application
has Owner	dApplicationComponent -> dApplicationConnectedWithApplication -> dApplicationComponent
connected From Application	Connected to Application
dApplicationComponent -> dApplicationAgregatesFunction -> dApplicationFunction	aggregates Function
part of Application	dApplicationComponent -> dApplicationComponentHasContainer -> dContainer
Has Container	contains Application
dBussinessProcess -> dProcessOrchestratesService -> dBussinessService	Has Service
part of Process	dBussinessProcess -> dProcessFlowsToProcess -> dBussinessProcess
Sequence Flow To	Sequence Flow From
dBussinessService -> dBizServUsesDataEntity -> dDataEntity	uses Data
is used by service	dBussinessService -> dRelatedService -> dBussinessService
Target Service	Source Service
dBussinessUseCase -> dBizServIsDescribedByUseCase -> dBussinessService	Has service
described by Use Case	dBussinessUseCase -> dUseCaseContainsActivities -> dActivity
has Activitiy	part of UseCase
dBussinessUseCase -> dFeatureIsRealizedByUseCase -> dFeature	realizes Feature
is realized by Use Case	dCapability -> dProcessEnsureCorrectOperationOfCapability -> dBussinessProcess
is operated by Process	Ensure the correct Operation of Capability
dCapability -> dCapabilityisMeasuredByKPI -> dMeasurementIndicator	is Measured By Indicator

Source→ Relationship Name → target	Source→ Relationship Name → target
is measuring Capability	dCapability -> dRoleExecutesCapability -> dRole
is executed by Role	executes Capability
dCapability -> dGoalisOperationalizedByCapability -> dGoal	support Goal
is Operationalized By Capability	dCapability -> dLogicalApplicationComponentSupportsCapability -> dLogicalAppComponent
supported by Application	supports Capability
dCapability -> dDataEntityInformsCapability -> dDataEntity	is informed by Data Entity
Informs Capability	dCluster -> dClusterContainsDeploymentNode -> dDeploymentNode
Contains DeploymentNode	is Contained in Cluster
dContainer -> dContainerAggregatesVolume -> dVolume	Aggregates Volume
is Aggregated in Container	dController -> dControllerImplementsFunction -> dApplicationFunction
Implements function	has Controller
dController -> dControllerGovernsView -> dView	Governs view
is Governed by Controller	dController -> dControllerControlsModel -> dModelClass
Controls Model	is controlled by Controller
dController -> dUseCaseisImplementedByController -> dBusinessUseCase	implements Use Case
Is implemented by Controller	dController -> dActionKey -> dController
target Controller	Source Controller
dDataEntity -> dEntityAssociatesEntity -> dDataEntity	Is connected to Source Entity
Is connected to Target Entity	dDecision -> dInitiativeisMakingUseOfDecision -> dInitiative
is used in Initiative	Makes Use Of Decision
dDecision -> dDecisionRefersToPrinciple -> dPrinciple	refers To Principle

Source→ Relationship Name → target	Source→ Relationship Name → target
is referenced by Decision	dDeploymentModel -> dDeploymentModelAggregatesCluster -> dCluster
Aggregates Cluster	is Aggregated in Deployment Model
dDeploymentNode -> dApplicationIRunsInNode -> dApplicationComponent	Runs Application
is run By Node	dEvent -> dEventTriggersProcess -> dBusinessProcess
triggers Process	is triggered by Event
dFeature -> dRequirementIsSatisfiedByFeature -> dRequirement	realizes Requirement
is realized by Feature	dGoal -> dGoalhasObjective -> dObjective
Has Objective	is related to Goal
dGoal -> dGoalAggregatesGoal -> dGoal	parent Goal
Child Goal	dGrowthPackage -> dPackageIsDeliveredByInitiative -> dInitiative
is delivered by Initiative	is part of a Package
dInitiative -> dInitiativeIncreasesMaturityOf -> dCapability	Increases Maturity Of Capability
is matured by Initiative	dIssue -> dIssueStopsRealizationOfRequirement -> dRequirement
stop Requirement	is stopped by Issue
dIssue -> dRiskRealizedByIssue -> dRisk	is realizing Risk
is realized by Issue	dJSON_Element -> dJSONElement_associates_JSONElement -> dJSON_Element
target JsonElement	Source JsonElement
dJSON_Schema -> dPolymorphicCollection -> dJSON_Schema	@baseType
@Type	dJSON_Schema -> dJSON_SchemaSubSetGeneralizes_JSON_Schema -> dJSON_SchemaSubSet
Parent Schema	Child Schema
dJSON_Schema -> dSchema_Associates_Schema -> dJSON_Schema	Target Schema

Source → Relationship Name → target	Source → Relationship Name → target
Source Schema	dJSON_Schema -> dSchemaGeneralizesSchema -> dJSON_Schema
Child Schema	Parent Schema
dJSON_Schema -> dSchemaAssociatesElement -> dJSON_Element	Associates Element
Associates Schema	dJSON_Schema -> dJSONSchemaRealizesModel -> dModelClass
realizes Model Class	is realized by Schema
dLocation -> dLocationHostsNode -> dDeploymentNode	Hosts Node
Is hosted in Location	dLogicalAppComponent -> dLogAppCompExposesBizService -> dBusinessService
application runs service	service run by application
dMeasurementArea -> dAreaAggregatesCategories -> dMeasurementCategory	aggregates categories
has Measurement Area	dMeasurementCategory -> dCategoryaggregatesGroups -> dMeasurementGrouping
aggregates Groups	has Category
dMeasurementGrouping -> dGroupAggregatesKPI -> dMeasurementIndicator	aggregates KPI
part of a Grouping	dModelClass -> dModelClassRealizesEntity -> dDataEntity
Realizes Entity	is realized by Class
dModelClass -> dTableProvidesPersistenceForModelClass -> dTable	is Saved in a Table
Provides Persistence For Model	dNetwork -> dNetworkContainsNode -> dDeploymentNode
Contains Node	is contained in a Network
dNetwork -> dNetworkHasSecurityGroup -> dSecurityGroup	Has Security Group
is associated to Network	dNetwork -> dNetworkAggregatesSubNetwork -> dSubNetwork
Aggregates Sub-Network	is Aggregated in Network
dObject -> dObjectInstanceofModel -> dModelClass	Has an Instance
Instance of Model	dObject -> dObjectInstanceofController -> dController

Source → Relationship Name → target	Source → Relationship Name → target
Has an Instance	Instance of Controller
dObject -> dObjectInstanceofView -> dView	Has an Instance
Instance of View	dOpinion -> dDiscord -> dOpinion
is in discord with other Opinion	is in discord with Original Opinion
dOpinion -> dAccord -> dOpinion	is in accord with other Opinion
is in accord with original Opinion	dOrganizationUnit -> dOrgUnitHasStakeholders -> dStakeholder
has Stakeholder	is concerned with Organization
dPhysicalService -> dPhysicalServiceImplementsBusinessService -> dBusinessService	Implement Biz Service
is Implemented by physical service	dProduct -> dProductIncludesProcess -> dBusinessProcess
Includes Process	Is part of Product
dProgram -> dProgramIsOrganizedInPackages -> dGrowthPackage	is organized In Packages
part of a Program	dRegion -> dRegionAggregatesCluster -> dCluster
Aggregates Cluster	is Aggregated in Region
dRequirement -> dRequirementIsRelatedToGoal -> dGoal	specifies Goal
is specified by Requirement	dRequirement -> dRequirementContainsRequirement -> dRequirement
is contained by Requirement	Contains Requirement
dResource -> dResourceSupportsCapability -> dCapability	Supports Capability
is supported By Resource	dRisk -> dRiskIRelatedToRisk -> dRisk
target Risk	Source Risk
dRisk -> dResourceimpactedByRisk -> dResource	impact Resouce
is impacted by Risk	dRisk -> dApplicationRisk -> dApplicationComponent

Source→ Relationship Name → target	Source→ Relationship Name → target
is related to Application	has Risk
dRole -> dRoleisAssociatedToOrganization -> dOrganizationUnit	works in Organization
has Role	dRole -> dRoleConsumesService -> dBusinessService
Consumes Service	is consumed by Role
dRole -> dRoleHasSkill -> dSkill	has SKill
known by Role	dSecurityGroup -> dSecurityGroupHasPhysicalService -> dPhysicalService
has Physical Service	Asociated to Sec. group
dStakeholder -> dStakeholderHasInnerOpinion -> dOpinionInner	Has Inner Opinion
is thinked by Stakeholder	dStakeholder -> dStakeholderHasOpinion -> dOpinion
Has Opinion	is expressed by Stakeholder
dSubNetwork -> dSubNetworkAggregatesIPRange -> dIPRange	Aggregates IP Range
is Aggregated in sub Net.	dTable -> dEntityisFunctionallyImplementedByTable -> dDataEntity
Implements Entity	is Functionally Implemented By Table
dTest -> dTest_ApplicationComponent -> dApplicationComponent	Test validates Component
is validated by test	dUserStory -> dUseCaseHasStory -> dBusinessUseCase
Is associated to Use Case	Can be told as User Story
dValueStream -> dValueStreamIncludesProcesses -> dBusinessProcess	Includes Process
is part of Value Stream	dValueStream -> dValueStreamEnablesCapability -> dCapability
Enables Capability	is Enabled by Value Stream
dZone -> dZoneAggregatesRegion -> dRegion	Aggregates Region
is Aggregated in Region	

6.3 Visual Characteristics of some DAF stereotypes

Some DAF stereotypes have special visual characteristics. They can react to Tagged Values or other properties, e.g. diagram properties as shown below.

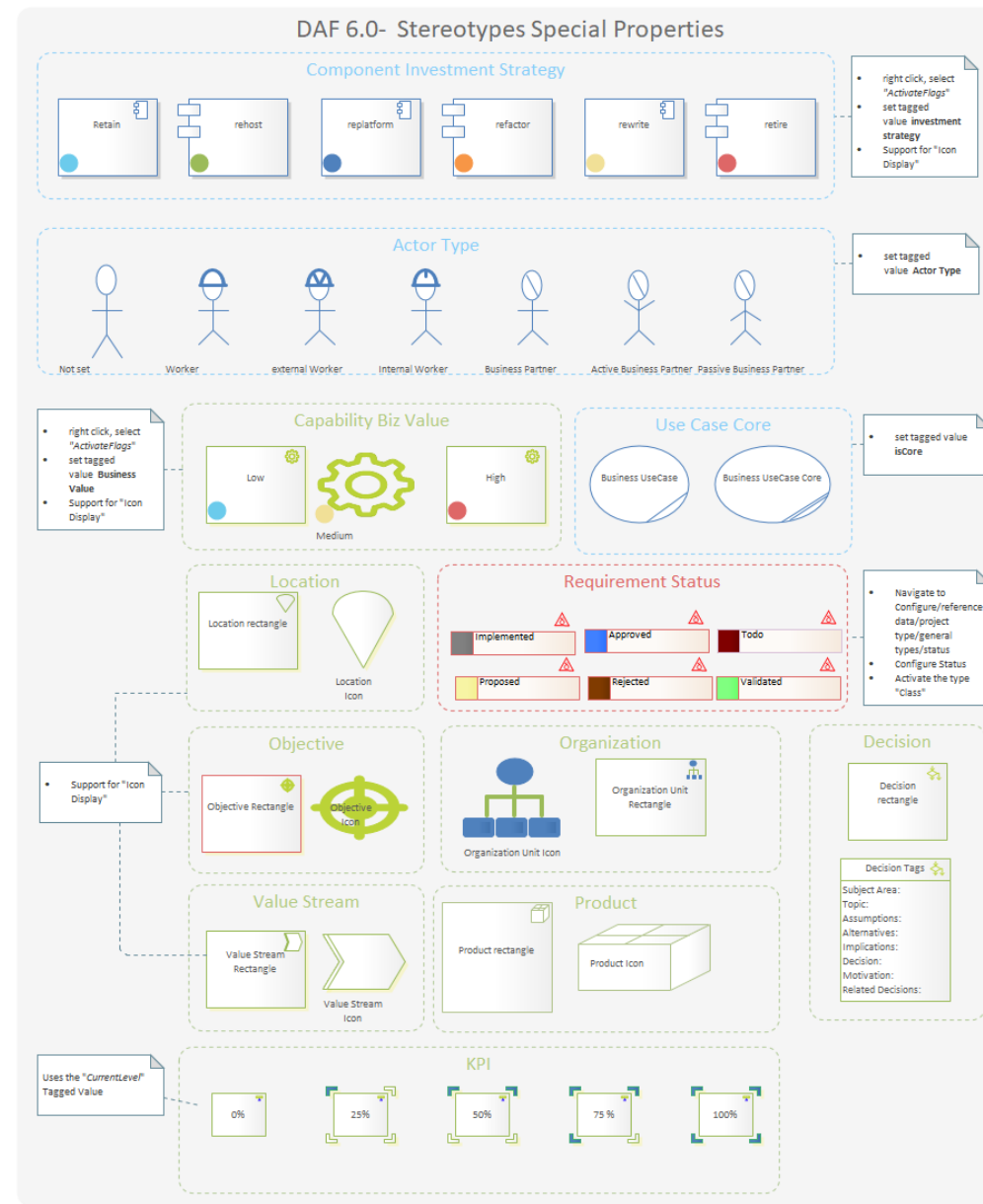
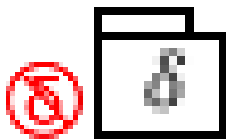


Figure 18: DAF Visual Properties

7 Appendices

7.1 Stereotypes and Tagged Values

Stereotype	Tagged Values	Notes
dApplicationFunction		a logical Function that the application is supporting
dRisk		the potential for an Issue
	Likelihood	the probability that this risk becomes an Issue.
	Severity	the expected impact if the risk materializes.
	RiskStrategy	the strategy to be adopted to manage the risk
dAction		A special  activity.
dActivity		A function  work carried out by the business user.
dActor		The actor  ty used for representing a User, System, a Worker or Partner.
	#FTEs	Estimated number of FTEs that operate as this Actor.
	ActorGoal	Objectives that this actor has, in general terms.
	ActorType	the type of Actor
	ActorTasks	Tasks that this actor performs, in general terms.
dApplicationComponent		An  of application functionality aligned to implementation structure which is modular and encapsulates its behavior and data, provides services, and makes them available through
	ID	Unique  for the element, useful for referring to third party systems
	Category	User-definable categorization taxonomy for each element.
	BizSatisfaction	How much is the business happy with the application, expressed in 1-100
	ITSatisfaction	how is IT happy with this application, often related to technical debt.



Stereotype	Tagged Values	Notes
	BizCriticality	how is Business happy with this application, often related to business debt.
	Cost	the cost of this application in 1000 of \$
	InvestmentStrategy	
	ApplicationType	the type of Application, e.g. if it's a COTS, Custom and so on
	LastStandardReviewDate	Last review date for the application component.
	NextStandardReviewDate	Next review date for the application component.
	RetireDate	The retire date for the Application Component.
	Owner	The owner for the application Component.
	Port	
	Source	The original source of the element's information
dBusinessProcess		A business  Processes represent a sequence of activities that together achieve a specified outcome, can be decomposed into Business Services. It ensures the correct operation of business capabilities.
	Category	User-definable categorization.
	ID	Unique identifier for the architecture entity.
	LastStandardReviewDate	Last date that the standard was reviewed.
	NextStandardReviewDate	Next date for the standard to be reviewed.
	Owner	Owner of the architecture entity.
	ProcessCriticality	Criticality of this process to business operations.
	ProcessType	Whether this process is supported by IT or is a manual process.
	ProcessVolumetrics	Data on frequency of process execution.
	RetireDate	Date when the standard was/will be retired.

Stereotype	Tagged Values	Notes
	Source	Location from where the information was collected.
	StandardCreationDate	If the product is a standard, when the standard was created.
	_metatype	
 BusinessService		1. A repeatable activity; a discrete behavior that a building block may be requested or otherwise triggered to perform. 2. A service provided by Business that achieve a business outcome in response to a request.
	Category	User-definable categorization for Business Service.
	ID	Unique identifier for Business Service.
	LastStandardReviewDate	Last date that the standard was reviewed.
	NextStandardReviewDate	Next date for the standard to be reviewed.
	Owner	Owner of the architecture entity.
	RetireDate	Date when the standard was/will be retired.
	Source	Location from where the information was collected.
BusinessUseCase		A Business Use Case is part of a business process that produces an advantage to the enterprise.
	GoalInContext	The goal should implicitly express the actor's intent or purpose of the use case, such as "Enrol Student in Seminar."
	Precondition	A list of the conditions, if any, that must be met before a use case may be invoked. Can be a previous Use case or self the presence of the system in Scope.
	Trigger	Event that is responsible for invocation of the use case.
	Scope	Boundaries in which the use case is operated when invoked (E.g. CMS)
	Level	Authorizations for operations/actions to be performed against the Chi business objects in scope. Against every object/process 4 CRUD basic operations are possible: Create (Write) Read (Open) Update (Change)

Stereotype	Tagged Values	Notes
		Delete (Destroy)
	OtherActors	The list of actors associated with the use case. Although this information is contained in the use case itself, it helps to increase the understandability of the use case when the diagram is unavailable.
	MainSuccessScenario	The main path of logic an actor follows through a use case. Often referred to as the "happy path" or the "main path" because it describes how the use case works when everything works as it normally should.
	Extensions	
	isCore	
 dCapability		A business outcome that is current available in an Organization Unit (or) A business-focused outcome that is delivered by the completion of one or more work packages.
	BusinessValue	assess how this capability provides value to the enterprise.
	Category	User-definable categorization for Capability
	ID	Unique identifier for the Capability
	Owner	Owner of the architecture entity.
	Source	Source from where the information was collected.
	Increments	Current (AS-IS) maturity level for the entity. 0- none 1- Initial 2- Under Development 3 - Defined 4- Managed 5 - Measured
	IncrementsToBe	Future (To-Be) maturity level for the entity.
	IncrementVertical	The maturity level of your vertical (the industry)
	IncrementSupplyChain	The maturity level of your supply chain (Partners)
	Cost	the aggregated cost of this Capability. Includes all the aggregated cost's elements

Stereotype	Tagged Values	Notes
	Criticality	What is the criticality of this capability, what would be the cost of failure
	Risk	How much Risk connected to this business Capability. 1 low risk, 100 max risk.
	CapabilityLevel	the level of this category in the hierarchy
 dController		The Controller defines application behavior and the Business Logic involving more than one ModelClass. It dispatches requests and selects views for presentation. It interprets user inputs and maps them into actions. Controller, with Views and Associations define the application flow. A controller represents the Business logic where a certain flow is physically implemented. In a Service Oriented Architecture (SOA) a Controller can be exposed by a Physical Service.
 dDataEntity		An encapsulation of data that is recognized by a business domain expert as a thing. Logical data entities can be tied to applications, repositories, and services and may be structured according to implementation considerations.
	PrivacyClassification	Level of restriction placed on access to the data.
	RetentionClassification	Level of retention to be placed on the data.
	Category	User-definable categorization.
	ID	Unique identifier for the Data Entity.
	Owner	Owner of the Data Entity.
	Source	Location from where the information was collected.
dDecision 		A decision-making step with accompanying decision logic used to determine execution approach for a process or to ensure that a process complies with governance criteria. <i>documents important decisions about any aspect of the initiative including the structure of the system, the provision and allocation of function, the contextual fitness of the system and adherence to standards.</i>
	Subject Area	Area of Concern
	Topic	Topic of Interest
	Assumptions	What is believed to be true about the context of the problem, constraints on the solution.

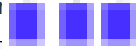
Stereotype	Tagged Values	Notes
	Motivation	Why this decision is important.
	Alternatives	A list of alternatives and explanations
	Decision	The decision taken, possibly with references to related work products
	Justification	Why the decision was made and a list of compliance to Architecture Principles and explanations of deviations from compliance.
	Implications	What impact the decision will have
	RelatedDecisions	A list of related decisions
dDeploymentNode 		a deployment node represents a physical or a virtual machine.
	IP	The Ip of this machine
	PublicIP	
	CPU	the number and type of CPU
	RAM	the Memory dimension
	Owner	The owner / responsible for this node
	Disk	the dimension of the disk
	ArchitectureType	the type of CPU Architecture (e.g. AMD 64)
dFeature 		A Feature is a fact that set his related Requirement to be true
	Author	This feature's author's name and role in the project
	Proofreader	This feature's revisor's name and role in the project
	Status	This feature's status
dGoal 		A measurable scope that the enterprise wants to achieve. Can be hierarchically decomposed.

Stereotype	Tagged Values	Notes
	ID	Unique identifier for the Goal.
	Value_amount	The actual amount of the value this goal intends to alter
	Value_Goal	The amount by which the value is to be altered
	Priority	A priority in %
	Value_Name	The name of the value this goal intends to alter
	dAssumption	The assumptions made to achieve the goal.
dGrowthPackage		A set of actions identified to achieve one or more objectives for the business. A work package can be a part of an Initiative, a complete project, or a part of a Program. A Growth pack is a collection of Initiatives architected for consistency and aimed to be executed in the same period.
dInitiative 		An activity carried out by a Business Unit or organization to achieve a particular goal or improve the maturity of a certain Capability.
	DetailedDescription	Detailed Description of the initiative.
	Impacted Capability	Describes the contribution this work package makes to capability delivery.
	RelatedProgram	The Growth Package or Program associated to the Initiative.
	startDate	The start date for the initiative.
	finishDate	The finish date of the initiative.
	InitiativeDuration	The expected duration of the initiative in Months/ Weeks or Years.
	kind	To capture the type of initiative.
	purpose	Purpose of the initiative.
	Rank	The rank of the initiative.
	Description	Description for the initiative.
	initiativeStatus	a set of colors to Visually describe the health of the initiative.

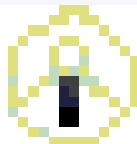
Stereotype	Tagged Values	Notes
dIssue		A concrete problem that is affecting a business. Also, a fact that sets a Requirement to be false.
	Author	This issue's author's name and role in the project
	Responsible	The responsible to close the present issue
dLocation		A place where business activity takes place. Can be hierarchically decomposed.
	AreaCode	The area code of the Location.
	City	The City of the Location.
	Country	The Country of the Location.
	EmailID	The contact e-mail Id for the Location.
	PhoneNumber	The contact phone number for the Location.
	Province	The Province of the Location.
	Street	The street part of the Location.
	ID	Unique identifier for the Location
dEvent		An organizational state change that triggers a business process; may originate from inside or outside the organization and may be resolved inside or outside the organization.
	Category	User-definable categorization.
	ID	Unique identifier for the Event.
	Owner	Owner of the Event.
	Source	Location from where the information was collected.
dLogicalAppComponent		An element of application functionality that is independent of a particular implementation.
dMeasurementArea		The organization or business units performance objective is captured as measurement area.
	ID	Unique identifier for the Measurement Area.

Stereotype	Tagged Values	Notes
	Definition	Definition of the Measurement Area.
dMeasurementCategory		The category to be measured under a measurement area.
	MeasurementArea	The reference Measurement Area to which this category belongs to.
	Definition	Definition of the measurement category.
	ID	Unique identifier for the Measurement Category.
dMeasurementGrouping		Categorization of measurement indicators into various groups under a measurement category.
	MeasurementCategory	Reference Measurement Category to which the group belongs to.
	Definition	Definition of the Measurement Grouping.
	ID	Unique identifier for the Measurement Grouping.
 dMeasurementIndicator		The specific measure captured for an organization or Business Unit which can be measured and quantifiable.
	ID	Unique identifier for the Measurement Indicator.
	Definition	Definition of the Measurement Indicator.
	UnitOfMeasure	The unit of measure (if available) for the Measurement Indicator.
	CurrentLevel	the level of the Measurement Indicator the last time that was measured.
	SatisfactionLevel	Satisfaction level for the Measurement Indicator, desired amount to be reached and/or Max value
	LastStandardReviewDate	last time this KPI was reviewed
dModelClass		Represent a class of the domain model, that can be converted into a Persistent domain class, must inherit from Node. It is the application's dynamic data structure
	child_order	The order of the associated children e.g. for Recipe: Image Info AdminInfo

Stereotype	Tagged Values	Notes
	display_value	The display value of a node e.g. for Recipe: name AdminInfo/status Info/status
	initparams	The configuration section that is used on initialization of the corresponding mapper.
	is_searchable	Indicates that this type should be included in the default search.
	is_soap	Indicates whenever an interface (e.g SOAP or REST) should be generated for this type.
	orderby	Definition of default sorting. Possible values: 'none' (no order), 'sortkey' (generates a 'sortkey' column, that is used for explicit sorting) or any the name of any dValue defined in the node optionally followed by [ASC DESC] e.g. 'name ASC'
	parent_order	The order of the associated parents.
	pk_name	The name of the primary key column. The generator will add this if there is no appropriate attribute.
	table_name	The name of the database table. If not given the name will be taken.
dObject		A dObject candidate for a dModelClass represented in an activity diagram. In alternative can be an instance of a Model Class
dObjective		A time-based milestone for an organization used to demonstrate progress towards a goal. The Objective is a non-final achievement, other than a Goal, the objective has no specific value to be measured
	Category	User-definable categorization.
	finishDate	the date in which the objective is achieved.
	ID	Unique identifier for the Objective.
	Owner	Owner for the Objective.
	Source	Location from where the information was collected.
dOpinion		A stakeholder view or judgment formed about a Topic. That is formulated in an explicit fashion
dOpinionInner		A Stakeholder's opinion that is relevant for the topic but is not formulated explicitly.

Stereotype	Tagged Values	Notes
dOrganizationUnit		 A s Org unit of resources with line management responsibility, goals, objectives, and measures. y include external parties and business partner organizations.
	HeadCount	 Nu working within the organization.
	ID	Unique identifier for the Organization Unit.
dPhysicalService		 the p lization of a Business Service
	Id	Unique identifier for Service.
	LastStandardReviewDate	Last date that the service was reviewed.
	NextStandardReviewDate	Next date for the service to be reviewed.
	Owner	Owner of the service.
	RetireDate	Date when the service was/will be retired.
	Source	information
 dPrinciple		A qualitative statement of intent that should be met by the architecture. Has at least a supporting rationale and a measure of importance.
	PrincipleID	Unique identifier for the Principle expressed as text.
	Implications	Statement of what the principle means in practical terms.
	PrincipleMeasurement	Identifies mechanisms that will be used to measure whether the principle has been met or not.
	Priority	Priority of this principle relative to other principles.
	Rationale	Statement of why the principle is required and the outcome to be reached.
	Statement	Statement of what the principle is.

Stereotype	Tagged Values	Notes
	Type	The category of the principle.
	Owner	Owner for the Principle
	Source	Location from where the information was collected.
dProduct		Output  ed by the business. The business product of the execution of a process.
	ID	Unique  r for the Product.
	Owner	Owner of the Product.
	Source	Location from where the information was collected.
	RetireDate	
	Price	
dProgram		a set of  ves that can be organized in Growth Packages
dRequirement		A business  ideline about the Enterprise or the project. A requirement is formulated in a SMART fashion and uses  row verbs. Requirements can be decomposed.
	ID	Unique identifier for the requirement.
	Author	This requirement's author's name and role in the project
	Status	the status of the Requirement in the workflow.
	Priority	A priority in %. Requirements are ordered by priority.
	Proofreader	Each requirement needs to be confirmed. This requirement's proofreader 's name and role in the project
	Type	The type of requirement.
dResource		a type  et supporting a business capability
	# of items	
dRole		The  ected function of an actor, or the part somebody or something plays in a particular action

Stereotype	Tagged Values	Notes
		or event. An actor may have a number of roles.
	#FTEs	Number of Full time Equivalent employees working in the Role.
	User	User-definable categorization.
	ID	Unique identifier for the Role.
	Owner	Owner of the Role.
	Source	Location from where the information was collected.
dStakeHolder		 Actor that has an interest in the considered domain. This is typically indicated by name, can be a Role in the organization. A stakeholder has open and Inner opinions that are relevant for the system.
	Legitimacy	Measure of general acceptance of this stakeholder influence
	Power	the measure of the possibilities in context that this person has to realize his objectives.
	Urgency	the timely importance of the Stakeholder opinions
dSystem		 A system or subsystem. Settings are modeled as attributes with default values
	config	The configuration file where the settings will be placed.
	platform	The platform to which the configuration settings apply.
dTable		 a physical table in a Database
	Database	The DB schema hosting the table
dUserStory		 a requirement expressed from the user point of view that achieve a business value
	TestScenario	The situation that verifies the user story is realized
	EstimatedEffort	Effort expressed in hours

Stereotype	Tagged Values	Notes
	StoryDetails	intention of the story expressed as "As a Actor I want to do something So that I achieve Value"
dValue		A Node type used in DAF. These values are persistent
	app_data_type	The datatype used in the application e.g. DATATYPE_DONTCARE, DATATYPE_ATTRIBUTE, DATATYPE_ELEMENT or DATATYPE_IGNORE
	db_data_type	The datatype used in the database
	is_editable	Whether the attribute is editable or not
	input_type	The HTML input type for the attribute e.g. select#fix:key1[val1] key2[val2]
	display_type	The HTML display type for the attribute e.g. image
	restrictions_match	A regular expression that the value must match (e.g. '[0-3][0-9]\.[0-1][0-9]\.[0-9][0-9][0-9][0-9]' for date values)
	restrictions_not_match	A regular expression that the value must NOT match
	restrictions_description	A text describing the restrictions
	column_name	The name of the database column. If not given the name will be taken
dValueRef		A Node type used in DAF that reference another Node Value, similar to a foreign Key. These values are persistent
dValueStream		A representation of an end-to-end collection of value-adding Business Processes that create an overall experience for the user, stakeholder or end user
	Criticality	
	is Decomposed	
	entrance criteria	
dView		A View type used in DAF. Controller, Views and Associations. It define the application flow from the user point of view.

7.2 Model Checks

Model checks are implemented using Bellekens EA Validator. See [Enterprise Architect Toolpack releases](#)

Following checks are implemented:

Category / Type	Description of the validation rules	Rationale
All		
	An Element with the status not set	A model element must have the status set
	Elements with the EA status Proposed not managed	Proposed status indicates that the Element is probably not managed
	The Main DAF diagram without objects	No diagram should be empty.
	A Diagram must not contain more than 25 elements	Ideally a diagram should not contain more than seven elements
	A Diagram should not contain less than 2 elements	Ideally a diagram should contain at least two elements
CIM		
Application Architecture	A "LogicalAppComponent" should exposes a "BusinessService"	a 'dLogicalAppComponent' element should be connected to a 'dBusinessService' element by a 'exposes Biz Service' Connector
Business Architecture	A Goal should be connected to a Business Capability	a 'dCapability' element should be connected to a 'dGoal' element by a 'isOperationalizedBy' Connector
	A Business Capability should be linked to a Measurement Indicator	a 'dCapability' element should be connected to a 'dMeasurementIndicator' element by a 'isMeasuredByf' Connector
	Logical Component is not linked to a Business Capability	a 'dCapability' element should be connected to a 'dLogicalAppComponent' element by a 'Supports Capability' Connector
	Missing link between the Business Process and a Business Capability	a 'dCapability' element should be connected to a 'dBusinessProcess' element by a 'Ensure the correct Operation' Connector
	A Business Capability is not supported by a role	a 'dCapability' element should be connected to a Role element
	Data Entity is not linked to a Business Capability	a 'dCapability' element should be connected to a Data Entity element

Category / Type	Description of the validation rules	Rationale
	An Initiative should support at least one Business Capability	a 'dCapability' element can be connected to a Initiative element
	A Business Capability is not linked to a Business Process	A Business process ensures the operation of a business capability
	A Business Capability is not supported by at least one Logical Application	A logical application should support at least one business capability
Requirement Management	A goal should be linked to a business requirement	a 'dRequirement' element should be connected to a 'dGoal' element by a 'is related to' Connector
	A Business Requirement should be linked to a Feature	a 'dFeature' element should be connected to a 'dRequirement' element by a 'isSatisfiedBy' Connector
Technology Architecture	The Network element should be connected to a DeploymentNode	a 'dNetwork' element should be connected to a 'dDeploymentNode' element by a 'NetworkContainsNode' Connector
	A Location element should be connected to a deployment Node	a 'dLocation' element should be connected to a 'dDeploymentNode' element by a 'Hosts Node' Connector
PIM		
Application Architecture	A Function has no documentation	A function should have some minimal documentation
	An ApplicationComponent documentation contains long dash '-' character	dApplicationComponent documentation should not have a long dash '-' character
	A dApplicationComponent does not have documentation	A dApplicationComponent should have some minimal documentation.
	An dApplicationComponent name should not contain 2 spaces between words in their name	An dApplicationComponent name should not have 2 spaces between words in their name
	An dApplicationComponent name contains leading or trailing space characters in their name	An dApplicationComponent name should not contains leading or trailing space characters in their name
	An dApplicationComponent documentation is in HTML format (rather than plain text)	An dApplicationComponent documentation should be in plain text not in the HTML format
	Application component has no name	Application component requires a name
	Function has no name	A function requires a name
	A function is not associated with an application	a function should be associated with an application

Category / Type	Description of the validation rules	Rationale
	An ApplicationComponent is not connected to an ApplicationFunction	a 'dApplicationComponent' element should be connected to a 'dApplicationFunction' element by a 'ApplicationAgregatesFunction' Connector
	An application Component doesn't realize a Logical Component	a 'dApplicationComponent' element should be connected to a 'dLogicalAppComponent' element by a 'Realizes' Connector
	A system is not connected to an ApplicationComponent	a 'dApplicationComponent' element should be connected to a 'dSystem' element by a 'configured by' Connector
	An applicationComponent does not exposes a PhysicalService	a 'dApplicationComponent' element should be connected to a 'dPhysicalService' element by a 'Exposes Service' Connector
	A PhysicalService should Implement a BusinessService	a 'dPhysicalService' element should be connected to a 'dBusinessService' element by a 'Implements' Connector
	A Controller should implement a Function	a 'dController' element should be connected to a 'dPhysicalService' element by a 'Implements Service' Connector
	A Controller should Control a ModelClass	a 'dController' element should be connected to a 'dModelClass' element by a 'Controls' Connector
	A Controller should Implement a BusinessUseCase	a 'dController' element should be connected to a 'dBusinessUseCase' element by a 'isImplementedBy' Connector
	A Controller should govern a View	a 'dController' element should be connected to a 'dView' element by a 'Governs' Connector
Business Architecture	A Feature is not realized by Business Use Case	a BusinessUseCase' element should be connected to a 'dFeature' element by a 'is realized by Use Case' Connector
	A BusinessService is not described by a BusinessUseCase	a 'dBusinessUseCase' element should be connected to a 'dBusinessService' element by a 'isDescribedBy' Connector
	Use Cases that are not in exactly one Boundary	A UseCase must appear in exactly 1 Boundary (of all Use Case-diagrams)
Technology Architecture	An ApplicationComponent is not physically hosted by a DeploymentNode	a 'dDeploymentNode' element should be connected to a 'dApplicationComponent' element by a 'isPhysicallyHostedBy' Connector
PSM		
Data Architecture	One or more Class diagrams has more than 25 elements	Update the invalid Class diagrams so they have maximum 25 elements
	There are some unreferenced Enumerations	Search and fix the Enumerations

Category / Type	Description of the validation rules	Rationale
	ere are some unreferenced Datatypes	All the data type must be described in a class
	Some Enumerations are without values	An Enumeration must have at least 1 value.
	A Model class has no name (or empty name)	Model class has no name (or empty name)
	An Association class is not related to other class(es)	Association should be related or connected to other class(es)
	The Attribute documentation is in HTML format rather than in plain text	The Attribute documentation should be in plain text and not any other format
	An Association documentation should not contain a long dash '-' character	Association documentation should not contain a long dash '-' character
	An Association documentation is in HTML format (rather than plain text)	Association documentation is in HTML format (rather than plain text)
	An Association name should not contain a long dash '-' character	Association name should not contain a long dash '-' character
	A Datatype/Enumeration documentation contain a long dash '-' character	Datatype/Enumeration documentation should not contain a long dash '-' character
	Datatype/Enumeration documentation should not be in the HTML format	Datatype/Enumeration documentation should be in the plain text format
	Datatype/Enumeration does not have any documentation	Datatype/Enumeration should have some documentation
	Some Classes containing Attributes are without datatypes	Every Attribute of a Class must have a Datatype
	Two classes with the same name in the model	Two classes with the same name cannot be in the same model
	A Model Class is without any relations to other entities	Entitie(s) are without any relations to other entities. This validation skips entities which are marked as <<baseType>>
	Model Class does not have Documentation	Model Class does not have Documentation
	Entity is directly derived from more than one base class	Entity is directly derived from more than one base class
	Entity is not directly derived from more than one base class	Entity should be directly derived from more than one base class
	Entities (classes or association classes) do not appear in any diagram in the model	Entities (classes or association classes) should appear at least in one diagram in the model

Category / Type	Description of the validation rules	Rationale
	An Attribute is not defined as public	Ensure that the attribute need to be != public
	An Attribute has no name (or empty name)	An Attribute must have a name
	Attribute of an entity does not have a type	Attribute of an entity should have a type
	Unnamed association (between 2 model Classes)	all associations must have a name
	Unnamed self-association	A self-association should have a name
	The Association name contains space characters	The Association name should not contain space characters
	Duplicate associations - associations with exactly the same name	associations must have unique names
	Model Classes are without Description	All Model Classes must have a Description.
	Association should specify navigability	unspecified navigability not recommended for information model
Implementation	An Initiative should be connected to at least one Business Capability	a 'dInitiative' element should be connected to a 'dCapability' element by a 'IncreasesMaturityOf' Connector

7.3 DAF COLORS

The metamodel uses certain colors to indicate domain of architecture

7.3.1 RGB Colors

- DAF Generic blue: 107,203,236
- Enterprise ARCHITECTURE Green: 155,187,89
- SOLUTION ARCHITECTURE dark blue: 79,129,189
- DESIGN ARCHITECTURE **Dark Orange: 233,191,149**

- Performance violet: 234,198,234
- Requirements red: 222,103,103
- Stakeholder light blue: 240,255,255
- Technology Green: 156,204,84
- Strategy Orange: 247,150,70
- COMMON yellow: 241,224,151

7.3.2 HEX Colors

- Blue: 107,203,236 → #6BCBEC
- - Green: 155,187,89 → #9BBB59
- - Dark Blue: 79,129,189 → #4F81BD
- - Orange: 247,150,70 → #F79646
- - Dark Orange: 233,191,149 → #E9BF95
- - Violet: 234,198,234 → #EAC6EA
- - Red: 222,103,103 → #DE6767
- - Light Blue: 240,255,255 → #F0FFFF
- - Green (another shade): 156,204,84 → #9CCC54
- - Yellow: 241,224,151 → #F1E097